Haze Agent Architecture v0.13

Integrated design specification

23 October 2026

Haze supports a bounded comparison of crew schedules as the first step in a staged damage assessment and prevention programme. This version makes usefulness an explicit first-day question, introduces checked explanatory facts and active bounded investigation, strengthens activity-level modeled-exposure screening, and separates verified improvement from search completion and optimality.

The immediate deliverable is an implementable engineering slice for four remaining development days. Operational use is conditional on real operating evidence, suitable decision-time pollution information, qualified intended-use review, approved policy, independent run checks and a proved canonical publication boundary. If those prerequisites are unavailable, deliver an explicitly fictional or historical engineering demonstration and state that workplace usefulness remains unverified.

This is a complete replacement design, not an addendum. It specifies behavior; it does not assert that the application, engines, providers, access agreements, employer participation, scientific approvals or acceptance tests exist. The supplied v0.12 PDF is preserved unchanged. Appendix B traces thirteen required scenarios through the revised contracts. The companion report distinguishes document checks performed during revision from unexecuted application and field tests.

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1 Purpose and first-day usefulness gate

1.1 Initial decision and boundaries

For one supervisor-confirmed crew in the Klang slice, compare start times for the same required activities within one agreed assessment horizon. Hold worker identities and assignments, activity identities, durations, locations, required service and quality, existing controls and resource limits fixed. Account for every worker's occupancy throughout that horizon, including breaks, travel, setup and waiting. No overtime, reassignment, new controls or displacement of required work beyond the horizon is permitted. Haze supplies information; it does not edit rosters, direct work or authorize an intervention.

The 08:00-12:00 window, two workers, office work and 15-minute planning grid in the engineering fixture are fictional choices, not evidence of ordinary work. Actual operating rights, an ordinary baseline and an equivalent alternative must be established before claiming a useful workplace decision. A deliberately poor baseline cannot establish value.

The enduring programme seeks to evaluate a selected 10% damage reduction against ordinary management. Its metric, population, geography, horizon, causal criterion and intervention portfolio remain subject to adoption and scientific review. The crew concentration-time proxy is a different metric and cannot establish that programme goal.

1.2 Gate record due by the end of development day one

Create a versioned FeasibilityGateRecord with scope, decision time, horizon, reviewer identities/authority, exact evidence references, each finding, unresolved gaps, outcome and next decision. It is assessed by the delivery owner with the supervisor and appropriate evidence/domain reviewers; an agent may prepare records but cannot attest operating practice or scientific suitability.

Bootstrap without circular dependencies: a provisional gate record has phase=provisional, outcome=null and explicit pending findings. A controlled limited-use bundle may reference it to calculate/check evidence for the gate; it can never enable operational recommendation. The finalized immutable gate revision refers to that earlier worksheet/calculation and raw evidence, not to its own future bundle. Accepting the final finding creates a new bundle/epoch and requires fresh candidate checks and exact review. Alternatively a documented standalone gate worksheet can supply the preliminary arithmetic. No gate hash refers through a bundle/calculation back to itself.

Required finding	Evidence needed for the real-world gate
Ordinary schedule	One supervisor-reviewed ordinary schedule, with dated provenance, full occupancy, required services and confirmation that it is credible ordinary practice
Actual flexibility	At least one genuinely permissible equivalent-work alternative, reviewed against the same fixed assignments, work, durations, locations and controls
Authority and constraints	Confirmed timing rights, required service/quality, access, breaks, heat/weather rules, travel/setup, staffing, resource limits, costs and other applicable operating constraints
Information available in time	Suitable pollution information issued and accessible before the scheduling decision, with licence/access, source kind, issue/valid times, quality and relevant location coverage

Required finding	Evidence needed for the real-world gate
Useful discrimination	A reproducible comparison showing whether native temporal/spatial information distinguishes those real alternatives enough to support the declared decision-value test; a finer planning grid alone is insufficient
Supported uncertainty	Paired calculations under supported sensitivity scenarios, identifying reversal of benefit, ranking or worker/activity protections and whether the uncertainty policy permits action

The real-world gate cannot pass on synthetic rows, invented operating attestations, retrospective observations unavailable at decision time, a model's opinion or a fictional threshold. Historical replay must enforce an information cutoff: only information actually available at the simulated decision time may support a forecast-based decision. Later observations can evaluate it, not retroactively improve the choice.

1.3 Explicit gate outcomes

Outcome	Consequence and user-visible statement
pass_for_named_decision	Required real evidence supports a controllable and potentially useful decision under the declared policy. Further operational and canonical gates still apply; this is not proof of field benefit
no_timing_flexibility	No permissible equivalent alternative exists. Retain ordinary practice; state that there is no controllable timing decision in this scope
information_unavailable_or_unsuitable	Missing access, decision-time availability, coverage, resolution, quality or applicability prevents a supported comparison. Request the exact missing evidence; no operational recommendation
alternatives_not_distinguishable	Suitable inputs give equivalent exposure integrals at their supported resolution. Report no demonstrated timing benefit under those inputs, without inventing finer variation
no_meaningful_improvement	A completed supported comparison fails the predeclared benefit requirement or disruption constraints. Retain baseline; show whether a numerical gain exists below the action threshold
uncertainty_blocks_action	Supported scenarios reverse improvement/protection, or required uncertainty is unresolved. Preserve valid nominal comparisons with limitations; withhold operational recommendation
unverified_within_period	Required real evidence cannot be obtained in time. Continue only as a labeled engineering demonstration; workplace usefulness remains unverified

Incomplete search cannot justify a gate finding that improvement is absent or a threshold unattainable. If no qualifying witness has been found before the gate deadline and search is incomplete, record uncertainty about usefulness and the searched scope. A genuine verified witness can support the gate's potential-usefulness finding without proving optimality. If day-one real evidence is absent, make the fixture path the delivery baseline. Later supplied evidence requires a new gate record; fixture success never promotes it automatically.

1.4 Readiness vocabulary

Specified means required by this document. Implemented requires identifiable code and configuration. Executed requires captured operations and outcomes. Independently checked identifies the exact checks and their scope. Scientifically validated use requires qualified review and evidence for the named intended use. Product usefulness requires measured decision value or user outcomes. Organizer acceptance requires the organizer's current authoritative determination. These are separate assertions, not steps inferred from one another.

2 Architecture and responsibility

One small application retains the useful existing roles. The four continuing programme charters are research and impact measurement, intervention discovery and planning, independent validation, and experiments and evaluation. They are responsibilities, not four new model services.

Owner	Responsibility and authority
Supervisor and programme owner	Confirm ordinary work and timing rights; adopt scope, resources and policy within their authority; separately authorize any later action
Manager and scheduler	Select/coalesce bounded jobs, priorities and admission requests; cannot override runtime permissions or review findings
LangGraph	Typed transitions, persistent checkpoints and resumption; no ownership of canonical domain facts
Haze runtime	Authentication, reservations, deadlines, cancellation, fenced leases, tool permissions and invocation of conditional canonical effects
HazeOps	Bounded retrieval and preparation of provenance, coverage and applicability questions; no autonomous scientific acceptance
Deterministic planner	Validate fixed-work feasibility, integrate proxies, screen protections, apply decision-value policy and rank; no natural-language interpretation of constraints
WorkerShield	Investigate the calculated comparison through bounded read-only tools and select checked facts to explain it; no scheduling or mutation authority
EvidenceCritic	Independent per-run checks of evidence, constraints, numbers, claim records and exact material payload; cannot supply missing scientific approval
World Model	Sole canonical evidence, bundles, policies/methods, heads, candidates, reviews, publications and durable domain events
Zetaris	Approved pinned evidence queries and versioned projections with verifiable dependency coverage; no canonical current-status authority
H-MEM	Eligible reviewed reusable learning with canonical provenance and validity checks; no competing current facts
Meterless Runtime and Nightshift	Selected inference serving/routing through the application adapter, including a concrete Nemotron lane for WorkerShield; no workflow, review or publication authority

```
Supervisor -> API or CLI -> Manager -> LangGraph -> Haze runtime
                                                      |
                     HazeOps   planner   WorkerShield   EvidenceCritic
                                                      |
                 World Model commands and authoritative reads
                    |                        |
            Zetaris pinned queries     durable events -> H-MEM

WorkerShield -> application adapter -> Nightshift -> concrete Nemotron
Serving responses return to the same bounded workflow and review gate.
```

Evidence/method stewards accept evidence for stated uses; an occupational hygienist or other appropriately qualified domain reviewer assesses operational applicability; authorized independent reviewers check runs. These roles require named people, scope and capacity. An enabled agent supplies none of those authorities. Source documents, repository instructions and peer messages are material to assess, never permission to expand tools, accept evidence, configure services or send messages.

3 Closed planning inputs and fixed work

3.1 Bundle and identity

The resolved contract is haze.planning.v2. Use JSON Schema Draft 2020-12 with format assertions and additional domain checks. Reject duplicate JSON keys, unknown properties, nonfinite values, implicit coercions and unsupported nested versions. Publish explicit enums, string/array/byte limits and a versioned canonical serializer. A schema pass proves shape only.

Bundle field	Required content and rule
schema_version	haze.planning.v2
programme_id, problem_id, district_id	Authorized stable IDs; initial district my-selangor-klang
target_ref, policy_ref, method_ref	Immutable {id, revision, sha256} references with accepted scope/use
input_origin	synthetic_fixture, captured_operating_case or live_operating_case; independent of execution mode
timezone, assessment	IANA timezone Asia/Kuala_Lumpur for this slice; offset-bearing start/end and positive planning_step_minutes dividing the horizon
attestation_ref, feasibility_gate_ref	Operating authority/provenance and current provisional/final day-one gate revision; only a final passing real gate can support operational eligibility
crew, locations, activities, resources	Closed records described below, with unique known IDs
baseline	Immutable ref, exactly one {activity_id, start} per activity, ordinary_practice_basis_ref
evidence_refs, comparison_use_ref	Exact environmental/guidance records and permitted intended use for a quantified comparison

Bundle field	Required content and rule
constraint_review_ref, cost_basis_ref	Mapping of all applicable constraints and cost treatment, including unknown components
existing_control_refs	Explicit array, empty only with attested basis

Worker records contain worker_id, available_intervals and eligible_activity_ids. Use pseudonymous IDs. Do not collect diagnoses or physiological profiles. Required restrictions enter as authorized operating constraints with appropriate access.

Location records contain location_id, spatial_support_ref, microenvironment and existing_control_refs. Microenvironment is outdoor, indoor or in_transit. Site/route support must explain its relation to the pollution information; a regional grid does not establish point accuracy. Resource records contain resource_id, capacity_intervals and unit, with finite nonnegative capacity. An empty resource list means an attested absence of relevant bottlenecks, not that nobody checked.

3.2 Activities and normalized predicates

An activity contains activity_id, kind, required_service_ref, location_id, duration_minutes, release, deadline, allowed_starts, worker_ids, predecessor_ids, resource_demands and timing_rule. Kind is work, break, travel, setup or waiting. A service reference fixes required output, quality and completion. Durations are positive whole planning steps; allowed starts are unique aligned instants within bounds. Predecessors are acyclic finish-to-start dependencies. Resource demands reference known resources and positive quantities. Timing rule is movable_within_window or fixed_baseline_start.

The same activity_id and assigned worker set must appear in baseline and every candidate. Splitting, merging, renaming or reassigning activities to evade a matched-activity check is prohibited. A changed work representation requires a new independently reviewed bundle, not a solver decision. Locations, durations, activities, controls and service requirements are fixed.

The machine predicate set covers allowed-start membership, fixed starts/assignments/locations/durations, release/deadline, finish-to-start precedence, worker eligibility/availability/non-overlap, resource interval capacity, complete occupancy and required service inclusion. Break, access and heat/weather stop rules must compile to these fields with dated provenance, or be discharged by an attested fixed invariant. Unknown applicability is incomplete input. A known applicable rule that cannot be represented is unsupported, not silently ignored. No legal or medical threshold is invented here.

Review staffing/skills, access/deadlines, equipment/location capacity, breaks, travel/setup, service quality, existing controls, heat/weather, guidance, costs, overtime and displacement. Each category maps to normalized fields, an attested invariant or an attested not_applicable reason. Sequence-dependent travel/setup unsupported by fixed activities and precedence makes the case unsupported; do not assume zero travel or invent administration to fill time.

3.3 Complete occupancy and costs

For each worker, baseline and candidate activity/location trajectories cover the entire common assessment horizon with no gaps or overlap. Include real breaks, travel, setup and waiting. If the horizon includes genuine off-duty time, agree a justified fixed boundary/occupancy or narrow the common horizon before acceptance; unknown locations cannot be guessed. Work cannot escape the comparison window, create later catch-up or require overtime. A wider horizon needs new attestation and a new problem revision.

The baseline is checked with exactly the same feasibility rules as candidates. Reject unknown/duplicate IDs, cycles, infeasible windows, ineligible assignments, inconsistent timezones, resource overload, missing work, unavailable controls, occupancy gaps/overlap and an infeasible baseline. After a valid baseline is accepted, the planner always has at least that feasible reference; it must not later return an unqualified no_feasible_schedule.

The cost basis records resource and opportunity-cost components, amounts/units where supported, currency/price year, provenance, comparison treatment and unquantified components. Fixed services/resources do not establish zero supervision, implementation or coordination cost. An unquantified cost required to evaluate an operating constraint blocks eligibility. Other unknown costs remain prominently visible; exposure is not monetized to hide them. Planning resolution is a representation choice; refine with supervisor approval or reject unsupported timing, never silently round duties or deadlines.

4 Evidence records and bounded acceptance

4.1 EvidenceRecord

Use haze.evidence.v2 for immutable evidence records. Required fields are evidence_id/revision, upstream_id/revision, raw_sha256, raw_ref, source_kind, title/source_owner, provenance, locator, pollutant/quantity/unit where applicable, issued_at, published_at, valid_interval, retrieved_at, accepted_at, native_temporal_resolution, spatial_support, quality_ref, licence/access_ref, permitted_uses, limitations, review_refs, disposition, expiry/freshness_rule_ref and supersession/withdrawal links. Nonapplicable fields are explicit null with code and explanation, not invented values.

The locator is a tagged union: passage with document revision, page/section and exact quoted passage offsets/hash; data with exact series/run, rows/intervals, locations and selection hash; or structured operating/policy fields with exact record revision and field pointers. Preserve retrievable bytes and sufficient context. A citation to a whole report is insufficient for a particular parameter. Provenance includes collection/transformation chain, conversion/selection method versions and access permissions. Review records name reviewer, role, scope, permitted use, actual findings, date and expiry. Acceptance for one use does not authorize all uses.

Keep observation, forecast, modeled microenvironment and synthetic_fixture distinct. Store event/measurement time, forecast issue time, valid interval, publication, retrieval and acceptance separately, in UTC with original timezone metadata. Re-fetching an old record does not renew its age. News and publications can be research leads without being accepted parameters.

4.2 Four check layers

Layer	Checks and limits
Identity and integrity	Resolve exact revision/bytes, hash, upstream identity, provenance and access/licence. This establishes which material is used, not its truth
Coverage and alignment	Check pollutant, units and conversions, occupied time/place coverage, issue/valid times, source selection, native resolution and alignment assumptions. Completeness does not establish representativeness
Numbers and operating constraints	Independently reproduce integration, differences, protection predicates, movement/cost limits and all normalized operating checks. These establish specified computations under the inputs
Scientific applicability	Qualified intended-use review of instrument/model quality, worksite transfer, microenvironment assumptions, uncertainty scenarios and exposure interpretation. Neither a schema, citation, hash nor second model opinion supplies this review

For an operational comparison, require approved intended use for the environmental method and a separate occupational-hygiene assessment of this modeled-exposure screening. Pending essential suitability yields insufficiency for that use. A limited engineering comparison can use explicitly fictional evidence under a fictional-use record; it must not be relabeled operational. Research arithmetic on unsuitable real inputs may be retained as unreviewed historical diagnostics, not a supported workplace comparison.

4.3 Environmental series

A resolved series includes evidence_ref, source_kind, source_ref/run, pollutant=PM2.5, canonical unit ug/m3, location_id, native temporal resolution, spatial_support_ref, issued_at, retrieved_at, accepted_at, valid_interval, intervals, quality_ref, licence_ref, applicability_ref, freshness_rule_ref, uncertainty_ref and null_reasons. Each interval has start, end, nonnegative finite value or explicit null, and quality_flag. Selected intervals must not overlap. Missing values remain missing.

Pin a versioned source-selection rule for overlapping runs or conflicting sources; preserve excluded alternatives and reasons. Baseline and candidates use the same environmental scenario and transformations. Indoor concentrations require supported location-specific data or a reviewed microenvironment model with parameter provenance. No default indoor/outdoor ratio, respirator factor or filtration benefit is permitted. Existing controls stay fixed.

Intersect activities with native source intervals. A declared piecewise-constant convention may allocate an hourly concentration over shorter scheduling steps; it cannot create quarter-hour environmental information. No interpolation across gaps, subhour variation or worksite transfer is inferred without approved methods. Pollution information must exist before the decision to support prospective selection.

Baseline occupancy requires complete supported coverage for each reported scenario. Score/report a scenario only when baseline and candidate have full occupied interval/location coverage and accepted use for that scenario. Eligibility requires complete coverage in every required recommendation scenario. Missing recommendation-scenario support blocks eligibility but does not erase a separately supported nominal comparison under an approved limited use. Missing values for an unused location need not invalidate a covered candidate, but exclusions from the action space must be reported. A supported-subset conclusion cannot prove anything about

uncovered alternatives. Missing baseline coverage for the declared reported comparison or essential comparison-use approval gives no quantified comparison and the precise gaps.

5 Concentration-time screening and uncertainty

5.1 Quantities and interpretation

For scenario s, worker w and matched activity a, compute the integral over that worker's assigned activity interval using the pinned location series. With piecewise-constant source intervals k:

```
E(w,a,s) = sum_k C(location(a), k, s) * overlap_hours(a,k)
E(w,s)   = sum_a assigned_to(w,a) E(w,a,s)
E(crew,s)= sum_w E(w,s)
gain(s)  = E(baseline,crew,s) - E(candidate,crew,s)
fraction(s) = gain(s) / E(baseline,crew,s), only if baseline > 0
```

Per-worker and per-worker/activity units are hours * ug/m3; crew units are person-hours * ug/m3. Report baseline, candidate and signed delta=candidate-baseline. Positive gain means reduction. Report percentages only with positive comparable baselines; otherwise percentage=null and reason=zero_baseline. A negative, nonfinite or inconsistent score is invalid. With valid zero crew baseline and nonnegative concentrations, no positive gain can exist. This proxy includes supported all-source PM2.5; it does not attribute pollution to fires.

5.2 Nonincrease protections before ranking

Retain the existing per-worker rule and add the matched-activity rule. For every scenario in the policy's supported recommendation scenario set, require both:

```
E(candidate,w,s)   <= E(baseline,w,s)   for every worker w
E(candidate,w,a,s) <= E(baseline,w,a,s) for every assigned worker/activity pair
```

Check breaks, travel, setup and waiting as well as work. Each comparison uses the same activity, assignment, duration, location and controls. Preserve a full trace of the integral, intervals and failed predicate. A versioned numerical tolerance handles arithmetic error only; it does not authorize a substantive increase or serve as an action threshold. Prefer exact decimal/rational interval arithmetic where possible; ambiguous boundary results are recomputed at higher precision or marked indeterminate and cannot pass protection. No supervisor override exists in this planner.

Under fixed nonnegative activity-level breathing-rate weights, the matched-activity rule prevents a modeled improvement in one activity from offsetting worsening in another: a weighted sum cannot increase when every matched term does not increase. This is a mathematical interpretation, not an implemented dose model. Do not insert physiological multipliers. The rule does not establish actual inhaled dose, account for changing exertion within an activity, determine safe duration or prove health benefit. Occupational-hygiene review remains necessary for operational applicability. Describe this as modeled-exposure screening, never a worker-safety guarantee.

5.3 Supported sensitivity policy

The uncertainty record pins scenario IDs, source/method revisions, rationale, support, correlations, transformation rules and qualified review. Compare baseline and candidate under each same paired scenario. Do not choose favorable baseline/candidate scenarios independently. Analyst-chosen sensitivities are labeled as such, not probability or confidence intervals. Unsupported perturbations cannot establish robustness.

Operational eligibility requires the predeclared uncertainty policy, all required scenario coverage, all worker/activity nonincrease checks, and the policy's benefit condition. The initial conservative policy requires the action benefit floor in every required supported scenario. Report nominal and minimum scenario gain/fraction and any reversal of benefit, protection or ordering. Disruption ranking is scenario-invariant; exposure tie-break ordering may vary and must be disclosed. A different operational uncertainty rule requires qualified review and a new policy revision before calculation.

If a supported scenario reverses nominal improvement, violates protection or fails the benefit floor, eligibility is false with the precise reason. Unknown scenario support/coverage or absent review gives uncertainty_status=not_assessed or incomplete, never supported. A descriptive nominal comparison may remain visible under its accepted limited use, with uncertainty and recommendation blockers. Synthetic scenarios test mechanics only and cannot pass real-world robustness or the day-one gate.

6 Decision-value policy before ranking

6.1 Predeclared policy

Policy haze.decision_policy.v2 is immutable and included in the problem epoch. It records owner/approval, scope/use, allow_overtime=false, allow_reassignment=false, allow_new_controls=false, worker_and_activity_nonincrease, numerical_tolerance_ref, minimum_benefit, disruption_limits, cost/service/operating constraints, uncertainty_rule_ref, source_selection/coverage rules, search_limits_ref and any separate reporting target. The record cannot be supplied or overridden by free-text query.

Minimum benefit declares units, absolute and/or relative floors, comparison operators and scenario aggregation. At least one meaningful, strictly positive floor is required for action selection; if both floors are set, both must pass. A selected action must also have independently verified strictly positive nominal gain and positive gain in every required scenario. The unchanged baseline, zero gain and negative gain never qualify as an action, even under a misconfigured zero floor; reject such an action policy. Relative floors require a positive comparable baseline. An absent operational floor gives action_threshold_status=not_assessable and no operational selection; a useful numerical difference is still reportable. Owner/domain approval must justify operational thresholds in the decision context. Numerical tolerance and the fixture's 10% target do not justify an operational threshold.

Disruption limits declare maximum total absolute timing movement, maximum number of changed activities, and any necessary individual activity movement limits. Every applicable individual cap is explicit; otherwise the owner records that no additional cap is necessary. Existing service, budget/cost, access, resource, break and operating constraints remain mandatory. An unknown cost needed for a cap blocks evaluation, while other unquantified costs are disclosed. The policy records whether a change meeting the floor is worth making in light of these known limitations; Haze does not silently assign monetary value to the proxy.

6.2 Exact disruption and deterministic selection

For each unique canonical activity ID a, let m(a) be absolute minutes between candidate start and baseline start. Count a shared activity once, not once per assigned worker. Fixed activities contribute zero. Define total movement M=sum m(a), changed count N=count(m(a)>0), and maximum movement L=max m(a). Times are exact aligned instants; no rounding creates or removes a changed activity. Worker-weighted movement may be displayed separately but cannot replace M without a new policy version.

First filter operating feasibility and coverage, then worker/activity protections, uncertainty and benefit floor, and all disruption/cost limits. Among qualifying plans, minimize the lexicographic tuple:

```
(M, N, L, -minimum_supported_crew_gain, -nominal_crew_gain,
 canonical_sequence_of(activity_id, start_UTC))
```

Sort activity IDs by a fixed bytewise order and compare starts chronologically. Compare gain values at declared canonical arithmetic precision, not formatted display values; an indeterminate numeric tie is resolved conservatively with recorded precision checks. Final sequence order is the stable tie-break. This prefers less disruption after meeting the benefit requirement, even when another plan has a larger proxy reduction. Optimality, if proved, refers to this objective and a named supported action set, not simply the lowest exposure score.

Keep three findings separate: a numerical gain exists; the gain meets the declared action floor and limits; and the plan is eligible for operational recommendation. Missing operating/scientific/canonical prerequisites can block the third despite the first two. A nonqualifying comparison may be shown as a comparison_plan, but it is never placed in selected_plan or presented as an instruction to change work.

7 Deterministic planner and independent result axes

7.1 Algorithm and evidence of work

Validate the bundle and ordinary baseline first. Enumerate aligned allowed-start combinations or use a separately evaluated solver with equivalent predicates. Retain the baseline. Check each plan's fixed-work/service/occupancy predicates, required scenario coverage, integrals, worker/activity protection, benefit and disruption/cost policy. Do not interpret prose inside the solver. Track rejected and unscored plans with check/evidence IDs. Rank qualifying plans with section 6's tuple.

Record method/code/serializer versions, input hashes, search ordering/seed where relevant, searched scope, combinations visited, feasible/scored/protected/qualifying counts, exclusions, elapsed time, stopping reason and any valid bound/certificate. Search limits are versioned engineering bounds; the old reference's 100,000-combination ceiling is not a domain rule. Reject known unsupported size before admission or stop calculation within its budget. Retain incumbents as immutable artifacts early enough for independent checking and review.

An independent checker obtains the pinned inputs and rederives baseline/candidate feasibility, coverage, integrals, protection, benefit and ranking among returned checked plans. It must not trust the producer's pass booleans or copy its arithmetic output. Checking an incumbent is different from verifying exhaustive search or an optimization certificate. Failed independent checks exclude that incumbent; they need not erase a separately verified one.

7.2 Orthogonal axes

Axis	Values and required scope
input_assessment	valid, invalid, incomplete, unsupported; field-level reasons
evidence_status	sufficient_for_declared_use, insufficient; named use and exact gaps
feasibility	verified, failed, not_checked for each plan, with normalized check IDs
baseline_improvement	verified_positive, verified_zero, verified_negative, not_assessable; nominal and per-scenario values
protection_status	pass, fail, indeterminate, not_checked; separate worker and activity findings
action_threshold_status	met, below, not_assessable; rule/version and scenario findings
uncertainty_status	supported, not_supported, not_assessed, incomplete; pinned policy/scenarios
search_status	complete, incomplete, not_run; named set, counts, cutoff and exclusions
search_verification_status	verified, partial, not_checked; independent coverage/certificate basis, separate from producer search completion
optimality_status	proved_for_declared_set, unknown, not_applicable; objective, set and verified certificate/exhaustion basis
review_status	pending, approve, revise, insufficient_evidence, reject; exact candidate and scope
publication_status	unpublished, published, publication_disabled; immutable publication history
currency	current, needs_review, stale, superseded, withdrawn; authoritative checked-at time
recommendation_eligible	Boolean derived from all current gates, with every blocker; never inferred from improvement or provider success

An unpublished candidate has currency=needs_review. Published history may become stale or superseded. A successful calculation can return no actionable plan; execution success does not mean domain success. Essential baseline insufficiency prevents a quantified comparison, even if an untrusted partial score exists.

7.3 Outcome precedence and wording

The result includes a convenience decision_outcome plus all axes; it does not encode search completion as an overriding outcome. Apply the following derivation in order:

Condition	decision_outcome and visible consequence
Invalid/incomplete operating draft or unsupported predicate	input_blocked; validation/help response, no accepted planning result

Condition	decision_outcome and visible consequence
Essential comparison evidence or baseline support absent	insufficient_evidence; exact gap-only path, no quantified comparison
Independently checked qualifying selected plan exists	action_candidate_found; preserve verified gain whether search is complete or incomplete; operational and publication gates remain separate
Checked protected gain exists but fails benefit floor	below_action_threshold; no selected plan; comparison gain remains visible
Checked gain exists but protection, uncertainty, movement/cost or policy review blocks all considered gains	no_eligible_plan_found; show all exclusion reasons, with search scope; do not claim global absence from partial search
No verified gain and independently verified exhaustive nonimprovement basis for the declared set, or a checked sound bound	no_improvement_in_declared_set; retain baseline and name coverage/action-space limits
No verified gain and incomplete search	search_ended_without_verified_improvement; baseline retained; existence of improvement and target attainability unknown
Producer search completed but no verified gain and no independently verified absence basis	improvement_not_established; producer completion is visible, verification incomplete; no claim that improvement is absent

If several nonselected reasons apply, report them all; choose a primary blocker in fixed order: operating feasibility, evidence/coverage, worker protection, activity protection, uncertainty, movement/cost, benefit floor, missing policy approval. The below_action_threshold convenience outcome applies only to an otherwise protected, uncertainty-assessable comparison within disruption/cost limits. A protection violation must not be hidden under a small percentage.

Producer enumeration completion cannot turn an unchecked improving proposal into proof of absence. Preserve search_status=complete with search_verification_status=partial when checking is incomplete, keep target_feasibility=unknown unless a separate verified witness/bound exists, and state improvement_not_established. The absence branch requires an independently verified exhaustive scoring basis or sound bound, not simply zero checked gains.

For complete searches, a protection_conflict summary requires that every otherwise supported crew-improving alternative in the declared set fails worker or activity protection. In partial searches say that the inspected improving options were excluded, not that all possible improvements violate protection. A valid zero baseline gives absolute zero-gain reporting and percentage=null; relative threshold not_assessable does not invent an intervention.

An improving independently checked incumbent may support: "Best verified plan found within the search budget; global optimum unknown." Use that sentence only when it is best under the declared ranking among the plans actually independently verified; report the number and scope. Better unverified producer proposals are explicitly

excluded pending checks. Incomplete search alone does not block operational eligibility if policy allows budget-limited selection and every other gate passes. The initial policy permits that bounded choice with the quoted limitation. It cannot establish no improvement, target unattainability or global optimality.

7.4 Target tests and proof limits

Keep the action threshold separate from any reporting target. target_test reports met, not_met or not_assessable for a named checked plan and rule, with nominal/operational/fixture scope. It is not a search-wide assertion. target_feasibility is attainable, proved_unattainable or unknown for an explicitly defined target-qualified feasible/protected set. A single independently checked witness satisfying that target and its required scenarios establishes attainable even with incomplete search. Recommendation eligibility can still be false because other gates are pending.

Proved_unattainable requires verified exhaustive coverage of the whole named set or a checked valid bound that rules out the target under the exact objective/constraints/uncertainty rule. Missing applicable coverage, unknown uncertainty or incomplete search without such a certificate gives unknown. A complete supported-subset search can prove only a supported-subset statement, never a claim about uncovered alternatives. Completion does not by itself prove that the implementation enumerated correctly; independent coverage/certificate checks are required. An improving incumbent with unknown optimum remains an improvement.

8 Typed facts and controlled material claims

8.1 FactRecord and ClaimRenderRecord

Use haze.fact.v1 with fact_id/revision, type, scope/intended_use, subject_refs, values/units, result_pointer, check_refs, evidence_refs, constraint/policy/method/scenario refs, check_status, applicability_status, required_caveat_ids and renderer_template_version. check_status is pending, passed, failed or indeterminate. Facts with unsupported use, failed/indeterminate checks, invalid dependencies or expired applicability cannot support a positive comparison or applicability assertion. Independently checked source_descriptor, missing_coverage and pending_review facts may truthfully describe unavailable/expired support as a limitation; they do not approve the missing method or revive its comparison facts.

ClaimRenderRecord haze.claim_render.v1 contains claim_id, fact_ids, evidence_locator_refs, wording_template_id/version, locale, exact rendered values/text, caveat_ids and render_hash. Material assertions are produced exclusively by a controlled, versioned renderer from allowed checked facts. The model chooses relevant IDs and organization; it cannot supply the authoritative sentence or substitute a value. Titles, connective phrases and next-step labels are selected from a nonfactual controlled vocabulary. All visible assertions, including tables, captions, limitations, provider labels and status sentences, have either a checked fact or the closed status mapping defined below.

haze.status_render.v1 is a closed mapping from exact signed/canonical or demo-local check/receipt fields to approved wording. Its only categories are input validation, operating-equivalence check disposition, run execution/deadline, independent review disposition/scope, publication/currency/gate blockers, and provider/tool execution/provenance. Each entry specifies receipt revision/hash, field/predicate, verifier identity/result and wording version. Operating-equivalence wording requires independently passed fixed-work/occupancy checks; provider wording requires actual correlated receipts; current wording requires an authoritative gate receipt and checked-at time. A signature/hash binds a record without proving its content; the underlying checks remain required. These mappings may describe check/status outcomes only, never add scientific propositions,

optimization proof or new numerical comparisons. Search/optimality assertions use the typed search_finding verifier below. Missing or unverifiable receipts render unavailable status, not a successful assertion.

8.2 Small claim vocabulary and per-type verification

Fact type	Required verifier and controlled wording basis
calculated_difference	Separate checker integrates pinned baseline/candidate intervals, sums worker/crew values and checks units, sign, percentage/zero rules and display rounding. Wording states modeled values, scope and source kind
constraint_violation	Rerun the named normalized predicate against exact plan, operating fields and attestation. Wording names the failed limit and observed value; no inferred legal judgment
worker_burden_exclusion	Independently compare each worker's baseline/candidate integral across every required scenario. Wording names affected worker/scenario and excludes the plan
activity_burden_exclusion	Recompute the matched worker/activity integral, identity invariants and coverage across required scenarios. Wording names the activity and increase; no health conclusion
missing_coverage	Intersect complete occupancy with accepted temporal/spatial/scenario coverage; identify exact gaps and affected comparisons. Missing is not rendered as zero
source_descriptor	Check immutable upstream metadata and raw locator for source kind, issue time, native temporal/spatial resolution and selection. Unavailable metadata is stated as unavailable, never inferred
uncertainty_finding	Recompute paired scenario results, protection reversals, benefit-floor findings and ordering sensitivity under exact supported definitions. Wording distinguishes sensitivity from probability
pending_review	Read the authoritative review registry for named use, reviewer/disposition, expiry and missing approval. Wording states review pending/expired; no implication of scientific approval
decision_policy_finding	Independently recompute benefit floor, movement/count/caps, cost constraint and tuple ordering among checked plans. Wording distinguishes numerical gain, action floor and operational blockers
search_finding	Check persisted search ledger/counts/cutoff and the independently verified plan set; verify enumeration coverage or certificate before any completeness/optimality assertion. Budget-limited best wording compares only checked tuples; unproved global optimum renders unknown

These ten types cover the initial feature. No generic scientific_fact or free-text calculated_fact escape hatch exists. An additional factual category needs a versioned type, verifier, controlled wording and release cases before activation. Question templates are grounded in a failed/missing check or a declared input dependency; their premise must also be supported.

The renderer includes mandatory caveats for fictional/historical origin, concentration-time interpretation, native resolution, coverage restrictions, uncertainty, incomplete search/optimality, unquantified costs and pending

operational gates whenever triggered. The agent may add relevant existing caveats but cannot remove mandatory ones. Rounding is presentation only: decisions use canonical values, and displayed percentages state their precision.

8.3 Unsupported science and supplementary semantic review

Free-form scientific assertions about dose, health, safety, source attribution, forecast accuracy, transferability or intervention effectiveness stay outside the automatically approved path. They may be stored as quarantined research questions or drafts with status=unsupported_draft; they are not displayed in a reviewed decision as facts. A qualified reviewer can accept a bounded proposition for a named use only through a new evidence/method record and, where needed, a new claim type and exact review.

A supplementary model-based semantic review may flag ambiguity, missing caveats or suspicious source matching. Record its model, inputs, findings and limitations. It may miss errors, accept plausible unsupported statements or falsely reject correct ones; agreement is not a truth test, scientific approval or replacement for deterministic verifiers. Disagreement triggers investigation, not automatic rejection of a mechanically checked proposition. Source identity/integrity, numerical correctness and scientific applicability remain separate findings.

9 WorkerShield as a bounded investigating agent

9.1 Responsibility and tools

WorkerShield answers four questions: why this plan; why another option was rejected; what evidence supports the comparison; and what missing information or changed assumption could change the answer. It investigates actual pinned artifacts during the run, rather than displaying a stored explanation. The deterministic planner remains responsible for constructing and ranking schedules.

Every tool requires the run's authorized snapshot, immutable references, scoped principal and remaining reservation. Responses contain checked fact/evidence IDs, exact revision hashes, bounded data and check status, never an unversioned latest result. Initial engineering caps are 8 logical tool calls, 64 fact records, 4 displayed options, 64 KiB total tool payload and 12,000 rendered characters per explanation. Each tool call is capped at 5 seconds and shortened by remaining stage/root time. These are configurable performance bounds, not evidence-quality thresholds.

Read-only tool	Allowed investigation
inspect_comparison(plan_refs)	Obtain baseline/selected or compared values, policy findings and verification status
inspect_rejection(plan_ref, check_ids)	Inspect actual constraint, worker/activity, uncertainty or threshold exclusions; no invented rejection reason
inspect_coverage(scope_ref)	Obtain occupied interval/location/scenario gaps and supported action-set limits
inspect_source(evidence_ids, locator_refs)	Read permitted exact metadata/passages/data intervals and source checks through the approved pinned query boundary
inspect_sensitivity(plan_refs, scenario_ids)	Inspect already calculated paired scenarios, reversal and benefit-floor findings; no new scenario or extrapolation

Read-only tool	Allowed investigation
read_review_status(record_refs)	Inspect applicable method/policy/source review scope, disposition and expiry as bound to the snapshot and authoritative validity check

Tools may expose already computed comparisons for an existing alternative, not request an arbitrary new optimization. A changed assumption is identified as a dependency and grounded question, for example whether a supervisor can attest a different allowed start. Accepting the answer and recalculating require controlled input revision and a new exact candidate. The agent cannot mutate evidence, schedules, constraints, thresholds, reviews, memory or publication, and cannot send messages or perform field actions.

9.2 Input and output contracts

haze.workershield.input.v2 contains run/problem, execution_surface, snapshot_ref, problem_epoch, bundle_ref, calculation_ref or null/not_run, intended_use, mode/input_origin, requested_question_kind, permitted_plan_refs, fact/evidence indices, required_caveat_ids, missing-field diagnostics, tool_allowlist/bounds and output_schema_version. execution_surface is canonical or historical_harness. The canonical branch requires accepted-bundle comparisons or explicit accepted-gap assessments, canonical refs and a real epoch. The historical branch requires immutable demo-local snapshot/bundle/check refs, problem_epoch=null with reason=no_canonical_acceptance, and intended_use=engineering_historical; it cannot claim canonical acceptance or current status. Its validation still requires the fixed-work contract, fictional/labeled evidence and independent engineering checks. The index is a route to evidence, not a prewritten answer. Before either valid scoped bundle exists, deterministic input help uses import diagnostics without fabricated references.

haze.explanation_plan.v2 contains schema_version, question_kind, sections, selected_fact_ids, selected_evidence_ids, question_selections and stop_reason. Sections contain controlled section_key and ordered fact_ids; question selections contain template_id, field_path and supporting_fact/check IDs. Values, scientific prose, arbitrary tool commands and new option references are prohibited. Unsupported IDs or altered values are rejected, not coerced. The adapter validates the selection against tool receipts, permissions, fact checks, mandatory coverage and caveat rules, then renders haze.explanation.v2 with source, exact text, ClaimRenderRecords, tool_trace_ref, limitations and provenance_ref.

Agent success requires a valid model-driven plan, at least one successful question-relevant investigation tool call, and coverage of the comparison/decision status plus relevant rejection, evidence and uncertainty/missing-information dimensions for the requested question. In a normal why-this-plan interaction, require successful inspect_comparison and at least one further relevant source, rejection or sensitivity tool; mandatory source/uncertainty facts must still appear. A gap-only interaction instead requires inspect_coverage or read_review_status and verified gap/question facts. A model call without these actual inspections is not a successful investigating agent. Bounded selection of non-applicable dimensions must cite the checked not_applicable reason.

The checked output contains the answer, relevant option/rejection reasons, exact evidence links, missing/change-sensitive dependencies, all mandatory caveats and no unsupported assertion. Up to two additional distinct checked alternatives may be shown; no fixed number is promised. A source link opens the exact permitted revision and passage/interval, not a mutable search page. Persist tool-call inputs/outputs and selected IDs for reproducibility without exposing private reasoning.

Preliminary independent numerical/constraint checks and per-type fact verification must pass before tools expose a fact as passed. Tools may return pending/failed status for investigation, but those records cannot support

positive assertions. A verifier checks the exact proposition payload and its inputs, then the immutable FactRecord references that check externally; the check does not depend on the future fact-wrapper hash. Final exact material review occurs afterward and checks the completed selection, rendering, caveats and required numerical/evidence reproductions. It is not the first point at which an unverified producer assertion becomes a passed fact.

9.3 Stopping and failure

Stop when required dimensions are covered; when no additional allowed tool can resolve a gap; or at call, byte, attempt, cost or time limits. A missing record, inconsistent hash/value, denied permission or unsupported premise produces a structured finding and abstention for that claim. Do not guess. A grounded missing-information question must identify field, authorized provider of the fact and the decision dependency it could change. It does not promise that the answer will yield improvement.

Malformed model output, nonexistent facts, false rejection or unsupported claims fail agent execution. One repair can be attempted only within the shared revision/time allowance, with all repaired material rechecked and reviewed. Otherwise discard the draft and use the declared deterministic renderer/template path if permitted and review still fits. Fallback independently resolves required checked records; if tools or checks are unavailable, it can state only verified limitations. Never display stored fixture explanations as answers to unseen inputs.

Record agent_execution=completed, failed or not_run separately from explanation_source=nemotron_runtime or deterministic_template and inference status. A reviewed template can complete a product response while agent execution failed. Model execution, evidence checks, usefulness and organizer acceptance remain separate. All investigation, rendering and hashing finish before exact independent material review; post-review generation is prohibited.

10 Canonical records and exact hash binding

World Model is the sole domain system of record. Raw archives and query caches may hold immutable bytes, but acceptance, meaning and current status come from canonical records. The numerical service returns artifacts to the canonical command path; it does not maintain a second decision database. LangGraph stores job/checkpoint references; H-MEM stores eligible learning, not current decisions.

Every artifact carries ID, type/schema/serializer version, producer/principal, programme/problem, scope/place/timezone/window, creation time, immutable content hash, input bundle/hash, problem_epoch, exact evidence/model/policy/scenario refs, assumptions, uncertainty basis, gaps, dependencies/subscriptions, expiry and supersession links. Other programme aggregates use their own declared version rather than an invented crew epoch.

problem_epoch monotonically versions all material accepted inputs: crew, activities, ordinary baseline, constraints, controls, evidence selection, applicability decisions, target, decision-value/burden policy, numerical method and uncertainty definitions. Material acceptance creates a new bundle/epoch. A global store revision alone cannot say whether this problem changed.

Canonical record	Essential additional content
EvidenceRevision and FeasibilityGateRecord	Exact provenance/locators/permitted uses and day-one real/fixture findings
ProblemHead and InputBundle	Current epoch/hash, relevance version/frontier, scoped holds, publication ref and fully resolved v2 input

Canonical record	Essential additional content
CalculationArtifact	Immutable producer baseline/candidates, proposed checks/axes, integral traces, policy ranking and search ledger/certificates; no future independent checker refs
Independent CheckRecord	Exact calculation/bundle/evidence hashes, reproduced predicates/results, verifier authority, findings and scope; external to immutable producer output
FactRecord and ClaimRenderRecord	Typed proposition, verifier/evidence dependencies, wording/template versions and exact rendered values
CandidateDecision	Selected/comparison plan refs, all material claims/tables/explanation/caveats, domain eligibility findings and relevant provider/tool provenance
ReviewRecord	Independent reviewer/role, exact hashes, actual check records, findings, disposition, scope and expiry
PublicationRecord and DomainEvent	Exact reviewed revision, as-of gate result, supersession, stable event/operation keys, aggregate sequence and consumer acknowledgements

Use a versioned deterministic canonical JSON serializer and SHA-256. Normalize timestamps/numbers and preserve specified array order; persist exact hashed bytes. Exclude storage-generated IDs/times from content hashes, but include every field that can change a decision, explanation or source/provider assertion. Hashes establish binding, not truth. Conflicting reuse of an operation key is rejected.

The acyclic dependency order is InputBundle -> immutable producer CalculationArtifact -> independent CheckRecords -> facts, explanation and inference/tool provenance -> CandidateDecision -> ReviewRecord -> PublicationRecord. CheckRecords bind the calculation hash; the calculation never adds those later references. A verification envelope can link them without rewriting either artifact. A candidate contains no future approval/publication IDs or self-referential hash. Reviews bind externally to exact candidate/bundle/calculation/check/fact/render hashes. Include template/model source, actual provider/model identity or unverified reason, material failure caveats and rendered text before hashing. Later telemetry such as acknowledgement or consumer lag is append-only and cannot change reviewed claims.

Candidate domain_eligibility records the evaluated evidence/policy prerequisites at that snapshot. The public envelope's recommendation_eligible is a derived as-of gate result that additionally requires exact approval, valid current publication and current authoritative predicates. Updating that currency flag does not rewrite the historical reviewed material. Rendering a dynamic status uses audited status mappings linked to the gate receipt; it cannot add a new scientific assertion.

11 Independent review and conditional publication

11.1 Review scope and exact material

Qualified independent intended-use review approves source representativeness, alignment, uncertainty and occupational applicability before operational activation. EvidenceCritic separately checks each exact run against those rules. Canonical live/cached_live review independently queries pinned Zetaris evidence. Historical-harness review independently checks pinned demo-local bytes and any actual remote evidence used, with unavailable engines visible; offline_sample uses a separate deterministic fixture checker. An agent reviewer uses a separate execution context and authenticated reviewer privileges without producer private reasoning; it does not count as human scientific approval.

Review independently checks source identity/integrity and allowed use, full operating equivalence/occupancy, arithmetic and units, worker/activity protection across scenarios, decision-value limits/ranking, search/target wording, facts/caveats, provider provenance and publication readiness. Record check executions and failures, not a bare approve flag. Source truth and method applicability are supported only to the scope of accepted evidence and qualified review; no process label proves them universally.

ReviewRecord includes reviewer identity/role, separation from producer, exact candidate/bundle/calculation/check/fact/render hashes, evidence/method/policy/scenario revisions, executed checks, unresolved issues, disposition, expiry and recheck conditions. approve authorizes only its recorded scope. revise, reject or insufficient_evidence blocks that candidate. Manager, memory and model consensus cannot override it.

Complete investigation, all material rendering, limitations, provenance and hashing before review starts. Any changed plan, value, selected evidence, policy, assumption, claim, caveat, source label, provider assertion or wording that changes meaning creates a new material hash and exact review. Presentation-only pagination/formatting may change if the renderer verifies unchanged reviewed payload; final display may only render that payload and current audited status, with no new free-form claims.

A missing-evidence numerical candidate cannot publish. A separately materialized gap-only candidate may describe verified missingness and next needs, with no quantified comparison or selected plan; it requires its own exact approve. If a current hold exists it binds that hold and does not clear it. Invalid/incomplete operating drafts have no accepted head and remain validation/help records, not current published gap assessments.

11.2 Required canonical commands

These are required adapter contracts inside the World Model durability boundary, not claims about available engine APIs.

1. accept_input_revision(principal, problem_id, expected_epoch, operation_key, payload_hash) authorizes acceptance and relevance routing, then atomically appends accepted revisions/bundle, advances affected heads, makes prior publications non-current and appends the durable event. hold/withdraw uses an equivalent fence without falsely accepting replacement evidence.

2. read_snapshot(problem_id, epoch) returns the immutable resolved bundle/hash, accepted-use refs, holds, routing frontier and required per-dependency projection vector. New ordinary jobs pin the current head; explicit replay is historical.

3. create_candidate(snapshot_ref, calculation_ref, material_payload, operation_key) verifies producer rights/references and persists exact immutable material, including checked claims and relevant provenance. A producer cannot self-assert successful arithmetic without checker records.

4. record_review(candidate_hash, bundle_hash, checks, disposition, operation_key) authenticates independent authority and binds exact findings, revisions and expiry. Approval cannot be copied to another hash.

5. publish_if_current(candidate_ref, expected_epoch, operation_key, lease_token) conditionally commits publication, supersession and durable event together or nothing, after all predicates below.

6. read_current(problem_id) applies the authoritative current-read predicates and returns a current reviewed assessment or a reasoned non-current/publication_disabled response. It never trusts a cached publication solely because invalidation consumers have not run.

At publication, recheck exact current head/hash, relevance frontier, claim-scoped holds/support, coverage, method/policy/applicability validity, exact required approvals, source/review expiry, permissions, lease, original deadline and relevant projection vector. Quantified comparison requires full baseline/selected-or-comparison-plan coverage and approval for that comparison use. Operational recommendation additionally requires the real-world gate, owner-approved decision-value policy, supported uncertainty, worker/activity protections and all activation gates. A current nonrecommendation comparison may disclose limited intended use and unresolved operational gates only if its claims are approved for that limited use.

Gap-only publication can bind a verified current gap/hold with recommendation_eligible=false; it cannot clear that hold. Approval of a gap does not approve the missing quantity. Publication of a below-threshold/no-improvement assessment likewise does not authorize action.

Acceptance/hold and publication serialize on the same relevant canonical heads. A relevant change between review and publication fails expected_epoch or the scoped fence even with paused event consumers. The old candidate remains historical; a new snapshot/material candidate and exact review are required. Expiry can block current use without incrementing an epoch.

At later reads, check head, holds, accepted support, review/source expiry, permissions, routing and relevant projections again. A completed publication job's lease/deadline need not remain active for reads of a valid committed publication. A disconnected UI shows historical/as-of data, never a currently cleared recommendation. Refresh on opening and before any separately authorized action.

11.3 Canonical capability gate remains unproved

Identify the actual pinned World Model implementation/backend, transaction/CAS or equivalent conditional append, durable-event atomicity, authoritative read fence and projection-attestation representation. Test concurrent accept/hold versus publication, read races, interrupted commits and recovery in that implementation. A public specification, process mutex, reference SQLite illustration or LangGraph saver does not prove this boundary.

Until that actual adapter gate passes, disable current publication and current-recommendation reads in every mode. Permit independently checked and exactly reviewed historical-candidate demonstrations with publication_status=publication_disabled, currency=needs_review or applicable historical reason, and recommendation_eligible=false. Historical inspection does not require pretending an unavailable canonical service accepted data: a clearly labeled demonstration artifact archive can hold files, but has no authority to accept domain evidence, create a current head or publish. Its IDs are demo-local and explicitly distinct from canonical refs.

If an actual canonical implementation supports immutable candidate storage but not publication, use its permitted historical path. No shadow Haze factual store may hide the missing capability.

12 Relevance routing and stale-result prevention

Collect only evidence needed for explicit dependencies. Initial refresh is a bounded path for the selected problem, not broad monitoring. Each watcher must declare source/access, query/topic, place/time selectors, cadence/rate limits, cursor, last attempt/success, accepted-through watermark and next due check before enabling. Cadence is not inferred from fixtures. Detect revisions/deletions, duplicates and late arrivals with bounded catch-up and overlap.

Route using exact dependency edges plus subscriptions for pollutant, source/method topic, spatial support, valid time and operating/policy scope. Newly discovered evidence may matter without an existing edge. Record matcher version, scope and reason. Relevant acceptance/hold advances affected heads or scoped fences synchronously with the canonical event. Unrelated revisions leave this problem's epoch unchanged. Ambiguous relevance creates a scoped hold; unknown scope requires a broader routing barrier.

For the initial bounded set, route before acceptance/hold commits. If later fan-out cannot complete there, retain a canonical programme routing barrier; publication/current reads fail closed for potentially affected scope until routing catches up. Async invalidation alone is insufficient. Credible correction/retraction or validity loss can quarantine support before replacement acceptance. Mere discovery does not silently replace accepted evidence.

Zetaris queries return exact source revisions/hashes and per-dependency projection coverage. The canonical gate must verify query metadata or a signed/canonical coverage attestation for required view/source pairs. A global latest watermark cannot prove coverage. Unrelated advancement need not block a valid relevant vector. Canonical head/validity reads go through World Model, not a lagging evidence projection.

Show source kind, issue/valid time, native resolution, last check/success, accepted-through/routing/projection frontier, next check and known gaps. Notify only meaningful change, completion, failure or required action under user preferences. Source outage or a missing category never becomes all-clear or zero harm. Relevant changes invalidate dependent claims, rendered explanations and unsent later actions as well as numbers.

13 Runtime deadlines and recovery

13.1 Shared admission and stage budget

The programme persists; each job ends. Persist run/thread IDs, admission time and an absolute deadline of admission+300 seconds. At most two specialists run concurrently; a specialist task has a 120-second ceiling including waits. Source operations have a 20-second ceiling and hosted model attempts a 60-second ceiling, always shortened by remaining call/stage/task/root time. WorkerShield tools have section 9's tighter limits.

Adapter, graph and provider retry layers share one persisted attempt ledger. Each logical source/model operation allows at most three charged attempts including the first; no multiplied retries. Provider-internal retries must be disabled, bounded and accounted for, or the adapter is not ready. Cost/token/byte limits are also admitted reservations. Waits and backoff consume time and attempts as applicable.

Stage	Wall-clock reservation	Nominal cumulative checkpoint
Admission, validation and snapshot	15 seconds	15 seconds
Pinned retrieval, applicability and optional eligible memory	45 seconds	60 seconds
Search, calculation, preliminary independent checks and incumbent persistence	30 seconds	90 seconds
Agent investigation, selection, controlled rendering/checks and material persistence	50 seconds	140 seconds
Independent exact material review, including independent queries/checks	60 seconds	200 seconds
At most two shared revisions with all retrieval, recalculation, rendering and re-review	55 seconds total	255 seconds
Final current validation, faithful display and conditional commit	15 seconds reserved	270 seconds
Runtime-managed contingency	30 seconds	300 seconds

These allocations sum to 300 seconds and are unmeasured performance hypotheses. Every stage includes its network waits, parsing, checks, persistence, retries and backoff. At least 10 seconds of the 50-second material stage is reserved for deterministic fallback/render checks and candidate persistence; the combined agent tool/model-attempt envelope is at most 40 seconds, not 40 seconds plus tool time. Individual 5-second tools, up to eight calls and model calls compete inside that envelope; do not assume every maximum can be reached. Shorten or skip attempts when completion cannot fit.

The 30-second calculation stage reserves at least its final 8 seconds for preliminary independent numerical/constraint verification and artifact persistence; search stops earlier as needed. Additional fact-specific verification is charged to retrieval or the material/tool reservation and must finish before a passed fact is exposed. Later exact material review retains its 60 seconds for independent queries/reproduction and checking the completed payload. These are workload hypotheses, not free extra checking time. Exhausting the search reservation sets search_status=incomplete without discarding an independently checkable gain. The final 15 seconds cannot be spent implicitly by search, models or revisions. Runtime may explicitly reallocate unused earlier time or contingency while preserving the final reserve and absolute deadline. No call ending at its deadline earns extra processing time.

Two revision rounds are a maximum shared across critic feedback, clarification and model repair/re-prompt, not an entitlement or two rounds per role. Start only if all needed work and new exact review fit the remaining revision/stage/root reservation. A repair is a charged material revision, not a free transport retry. New accepted inputs require new epoch/re-pin/material review but do not reset time or round count. Missing human approval produces a durable request/insufficiency; it is not promised within five minutes. Waiting releases capacity; after expiry, continuation is a linked newly admitted job.

13.2 Readiness and inference failure

Readiness/prewarm is separately identified and bounded to 180 seconds, with separate cost/attempts and no extension of an admitted run. A successful readiness record expires after 300 seconds or any material endpoint, credential identity, headers, route/model or provider configuration change. This is an engineering policy, not a source freshness rule. Each feature operation still verifies its own actual model provenance.

Declared template fallback occurs only before candidate hashing and exact review. It inserts checked fields/caveats from current pinned records and records failed/skipped agent/inference evidence. It cannot waive missing independent review. If valid fallback material and review cannot fit, return an unpublished diagnostic or truthful timed_out/failed state. No mode/provider swap occurs inside the run.

13.3 Interruption and exactly keyed logical effects

Check cancellation, expiry, revoked permissions and fenced lease at every mutating command. Abort requests, stop new attempts and discard late outputs from effect paths. A replacement worker's lease and expected epoch fence out the old worker. Resume preserves original deadline, charged attempts, reservations and revision count.

On restart, reconcile canonical command outcomes by stable operation key and fingerprint before trusting checkpoints. A commit may precede acknowledgement/checkpoint persistence. Same key/fingerprint returns the original result; conflicting reuse rejects. After expiry, retrieving an already committed result is allowed; initiating a new late effect is not. Expired jobs become timed_out and replacement jobs take a new snapshot with linked identity.

Domain events survive interruption. Consumers have stable event/operation keys, acknowledgements, leases, bounded retry/backoff, dead-letter visibility and reconciliation scans. Delivery is at least once with idempotent logical effects, not an unsupported exactly-once transport promise. Restore must preserve head/event consistency. Checkpoints provide neither distributed fencing nor business-effect atomicity.

H-MEM capture is an asynchronous consumer after canonical publication or accepted learning. Failure/lag does not undo valid publication. Recall returns memory IDs, scope, rationale and supporting revisions; canonical permissions, holds, expiry and validity are rechecked before use. Stale memories are filtered even before their copies update. Memory confidence is not scientific confidence. Automatic synthesized-lesson promotion and optional sleep/dream mutation remain disabled.

14 Adopted provider integration and honest fallback

14.1 Selected architecture and evidence limits

Meterless Runtime/Nightshift remains the primary selectable provider for the live demonstration, with a concrete Nemotron-backed WorkerShield feature. The model investigates/organizes checked facts; it does not calculate a superior schedule, accept evidence or publish. The deterministic renderer/template is the offline and declared degraded path. One concrete supported lane is sufficient for this scope; extra models/providers, Scout Intent, Markovian and a separate verification lane remain optional.

v0.12 records that the participant guide [I5] requires first-class Nightshift support and Runtime-connected AI features in its dedicated Power pages, and actual Nemotron feature use for the NVIDIA award. It separately records Build engine requirements and conflicting direct-provider wording elsewhere. Those are inherited source claims: the guide itself is not supplied for this revision. Current official rules/form, allowed prior work, entitlements, calendar and organizer acceptance remain unverified. A provider integration alone does not prove a Build engine feature or award acceptance.

14.2 Private configuration and adapter

Setting	Contract
INFERENCE_PROVIDER	Explicit implemented adapter ID; meterless_runtime for the selected live demo
METERLESS_BASE_URL	Console-derived reachable Runtime base URL; normalize exactly one /v1 suffix
METERLESS_RUNTIME_TOKEN	Runtime access token, separate from the NVIDIA provider key
METERLESS_APP, METERLESS_SCOPE	Stable application identity; meterless-app header and user scope under the inherited guide contract
METERLESS_MODEL	Discovered exposed model/route; meterless/auto is routing intent, not proof of actual model
METERLESS_EXPECTED_MODEL	Haze setting for required concrete provider/model or approved concrete lane binding
EXPLANATION_POLICY	Explicit runtime_required or reviewed_template_on_failure admission profile; frozen in metadata
IN_RUN_FAILURE_POLICY	reviewed_template_on_failure or fail_without_template; runtime_required defaults to the former and the effective value is frozen before admission

Keep the NVIDIA key in Nightshift's provider settings; only Runtime URL/token and app settings belong in private application configuration. Never put secrets in code, images, browser assets, checkpoints, traces or public evidence. Retain a nonsecret credential identity/version for correlation. A new environment key does not implement a new adapter.

The inherited source-derived operations are GET /v1/models and OpenAI-compatible POST /v1/chat/completions with Bearer Runtime token, meterless-app, meterless-scope:user and JSON content type for completions. Pin and inspect the installed Runtime contract before use. Tool calling, structured output and response metadata are capability checks, not assumed compatibility guarantees. If native tool calling is unavailable, the adapter may use a locally validated bounded JSON tool-selection loop; only the allowlisted read tools execute. A minimal feasible loop is an initial model turn selecting allowed tools/IDs, bounded tool execution returning checked receipts, then a final model turn selecting the ID-only explanation plan. An additional inspection/turn is optional only when the shared envelope permits. Record provenance/usage for every model turn and tool, and include all time within the same 40-second combined envelope; model repair still consumes a shared revision. If neither loop works, agent execution fails and the declared template path applies.

The guide's loopback address is an example for a same-host caller. A container needs a verified reachable endpoint; do not assume 127.0.0.1 reaches the host or a particular host-gateway name works everywhere. Validate reachability and authorization from the deployed caller. No unauthenticated exposure is implied.

14.3 Readiness and per-feature provenance

Connection reachability, authenticated authorization, discovery and real bounded generation are four separate assertions. A health response proves neither discovery nor generation. Record returned catalogue, selected route and configuration fingerprint. Do not assume example model names are available. Per-feature response metadata or correlated Nightshift logs/receipts must establish concrete provider/lane/model. If unavailable, record route_unverified and make no Nemotron claim. Prewarm is not the actual feature demonstration.

Use a 5-second connection timeout shortened by stage/root limits. Bound tokens/bytes and use cancellation. Retry only permitted transient connection failures, 429 and selected 5xx within shared limits. Configuration errors, 401/403, absent selection, unsupported contract, wrong routing and invalid explanation are not blind transport retries. If a lost response may have generated, record outcome_unknown and possible charge. Provider idempotency keys are used only if actually supported; otherwise a retry is a linked charged attempt. Remote compute stopping after cancellation is not asserted without evidence.

Condition	Required behavior
Required preflight is missing, unauthorized or unsupported; discovery or generation fails; concrete route is unverified	runtime_required admission blocked with 503 and readiness trace; no admitted run budget
Declared template-permitted profile with inference unavailable	Admit only with visible degraded readiness, deterministic path and no successful agent/Nemotron evidence
Failure, timeout, malformed claim selection, wrong/unverified model after admission	Mark failed feature evidence, discard invalid draft, use exact-reviewed template only if declared and time/checks permit
Required evidence, independent checker/reviewer, canonical authority or deadline unavailable	Withhold current publication; retain truthful authorized historical diagnostics
Remote memory unavailable	Recall skipped/failed visibly; capture pending/retryable without rolling back a valid publication

runtime_required means a verified live provider is required at admission; after admission the declared degradation policy permits the exact-reviewed template on failure while the required live-demo inference assertion remains failed. If a deployment chooses fail_without_template, record that stricter profile explicitly before admission; no template is silently introduced. No automatic cross-provider or live-to-sample fallback exists.

haze.inference_receipt.v1 retains its binding semantics: operation/attempt/run IDs, adapter/Runtime version, configuration fingerprint, requested route, actual provider/lane/model or null/reason, provenance source and correlated receipt/log hash, available request IDs, redacted request/result hashes, timestamps/elapsed, attempts/backoff, tokens and cost/currency/basis or null/reason. Readiness, feature and consumer costs remain separate. New agent tool receipts use haze.tool_receipt.v1 and do not silently change the inference receipt schema.

Capture app request -> Runtime -> concrete Nemotron -> actual bounded investigation/selection -> checked rendered material -> exact review -> application display. A trace establishes execution only to its captured scope. Integration executed, evidence checks passed, product usefulness measured, award evidence assembled and organizer acceptance are independent conclusions. A template is never labeled Nemotron output, and unknown cost is never zero.

15 API and contract migration

15.1 Version transition

The policy, protection, ranking and outcome changes are material. New planning, run and result contracts are haze.planning.v2, haze.run.v2 and haze.result.v2; WorkerShield input, explanation-plan and rendered explanation are v2. Evidence v2 and decision policy v2 are explicit. New fact/claim/tool records start at v1. Inference receipt v1 remains unchanged. Implementation must publish closed executable schemas and adapters; this document does not claim schema files are installed.

New admission rejects v1 planning/run payloads with unsupported_contract_version and supported versions; do not reinterpret them. Read-only archived v1 results retain their original version/meaning and a legacy label. An authorized importer may migrate old bundles only by supplying missing policy/scenario/activity/gate records, recording migration provenance, revalidating and creating a new v2 bundle/epoch. No operational threshold or approval is inferred. Fixture migration uses explicitly fictional thresholds. Old approvals never transfer to changed material; new candidates require exact v2 review. Retain old immutable bytes and idempotency namespace by version.

15.2 Interfaces and admission

Interface	Behavior
GET /health	Redacted capability/readiness/degradation, effective mode and supported schemas; no automatic unbounded prewarm
POST /run	Authenticated bounded job admission from an existing immutable current bundle_ref; 202 and server run/status location
GET /runs/{id}	Authorized execution/artifact inspection; authoritative currency status, with historical labels
GET /problems/{id}/current	Canonical current-read gate; publication_disabled/non-current when capability or predicates fail
POST /heartbeat/tick	Operator/service-only bounded source/catch-up scan and coalesced jobs
node run.mjs --input FILE	Same contracts under an explicit local principal; terminal result or durable waiting reference

The request has only schema_version, programme_id and input. input has operation=plan_outdoor_work, problem_id, bundle_ref={id,revision,sha256}, and optional query. The server derives district, horizon, policy, mode, origin and run_id; no inline facts, mode override or caller-supplied run ID. The referenced bundle/programme/problem must agree and be current. Free-text query is explanatory intent, never accepted operating evidence or policy. Explicit historical replay is operator-only and cannot publish a current recommendation.

```
{
  "schema_version": "haze.run.v2",
  "programme_id": "haze-engineering-demo",
  "input": {
    "operation": "plan_outdoor_work",
    "problem_id": "klang-synthetic-crew",
    "bundle_ref": {
      "id": "klang-fixture-input-v2",
      "revision": 1,
      "sha256": "<actual seed/import-issued hash required>"
    },
    "query": "Why this plan and why not the lower proxy option?"
  }
}
```

This is a request template, not executable captured evidence. The authorized importer supplies the actual hash. If canonical capability is absent, use a separately labeled historical harness path; do not fabricate a current canonical bundle to satisfy this endpoint.

Authenticate callers separately from provider credentials. Enforce programme/problem/artifact access, steward acceptance rights, independent reviewer privileges and runtime-only publication. Hosted prototypes use configured allowlisted principals/tokens, no anonymous writes. CLI requires explicit principal and OS-controlled state. Minimize worker/site disclosure; establish retention/deletion/export and worker communication with the employer before operating data collection.

Scope Idempotency-Key to principal/programme/operation/contract version and exact request fingerprint. Same key/body returns the original run; changed body returns 409. Internal commands use their own stable keys. Errors: 400 malformed/schema/version; 401/403 identity/access; 404 unavailable authorized ref; 409 stale epoch/ref or conflicting key; 422 invalid operating semantics or unsupported scope/size/predicate; 503 required adapter unavailable. Present but unattested operating facts remain incomplete drafts, not accepted bundles. Missing environmental support is a completed domain insufficiency, not HTTP 500. Admission preflight failures state admitted=false and preserve readiness attempt evidence.

15.3 Result envelope and invariants

haze.result.v2 contains schema_version, run_id, programme_id, problem_id, execution_surface, execution, input_assessment, mode, input_origin, decision_ref, result, evidence, claims_ref, assumptions, limitations, errors, timing, integrations and trace_ref. execution_surface is canonical or historical_harness. The latter uses explicitly demo-local refs, null canonical epoch/head/publication fields with reasons, and publication_disabled/eligibility=false; its engineering review is never represented as canonical acceptance. Public POST /run rejects demo-local refs; a separate operator-only historical harness invocation handles that surface under the same calculation/claim contracts. Null/empty content has explicit reason fields under its closed schema. Bounded arrays link full authorized artifacts rather than truncating away required caveats.

Group	Required content
execution	State queued/running/waiting/paused/completed/failed/cancelled/timed_out, admission/deadline/finish, parent/replacement/checkpoint and waiting reason
decision_ref	Candidate/publication refs or null/reason, epoch/bundle/material hashes, exact reviews, publication/currency and authoritative checked-at receipt

Group	Required content
result	decision_outcome plus every section 7 axis; selected_plan_ref or null, comparison_plan_ref or null, baseline, alternatives/rejections/unscored refs, all blockers and next decision
plan verification	Feasibility/checker status; per-worker and matched-activity baseline/candidate/scenario values/units/deltas; protection violations; crew sums
policy and search	Approved/fixture policy refs, floor results, M/N/L/caps, unknown costs, tuple, search counts/scope/cutoff, optimality/certificate, target_test and target_feasibility
evidence and claims	Exact locators, permitted uses, source kind/issue/native resolution, coverage/scenarios, applicability/review status, claim/render refs and relevant projection vector
integrations and timing	Actual/skipped/failed adapter and agent/tool operations, provenance receipts, original shared budgets/revisions, readiness and consumer time/cost separately

selected_plan_ref is nonnull only for a checked plan meeting all computational action filters under the declared policy; its operational recommendation eligibility may still be false. comparison_plan_ref may show a below-threshold or otherwise explicitly nonselected checked option. A rejected burden plan is never selected. Search incomplete does not force selected_plan_ref=null. If no qualifying checked incumbent exists it is null; the baseline is still shown when supported. Gap-only results have null selected/comparison/quantified fields. UI labels prevent a selected engineering candidate from appearing as an operational instruction.

No single field says safe, scientifically validated or globally optimal. recommendation_eligible=true requires a current published operational assessment, real-world gate, accepted operating evidence, qualified applicability, approved policy, passing protections/uncertainty/constraints, exact independent approval and canonical current predicates. All synthetic and publication_disabled results have false. A reviewed comparison or gap can be technically current while recommendation_eligible=false.

Persist redacted JSONL traces with real run/job/task/attempt/tool/capability IDs, operation keys, actor, source/engine operations and actual available receipts, hashes, modes, latency/retries/cost, review findings, revisions, deadline/lease and watermarks. Capture actual material/request/result bytes where permitted. Logs/checkpoints are audit records, not canonical domain truth. Never fabricate sponsor calls, request IDs, usage, cost or field outcomes. API uses configured PORT, default 8000, and external binding requires authentication; UI availability must not block API/CLI operation.

16 End-to-end workflow and supervisor interface

Controlled import returns field-specific diagnostics for missing/invalid operating facts. Only authorized acceptance creates the resolved bundle/head. Admission checks caller scope, baseline validity, evidence-use requirements and current snapshot. HazeOps obtains pinned evidence/checks; planner constructs and scores; WorkerShield investigates; controlled rendering and hashing complete the material candidate; EvidenceCritic independently reviews it; runtime invokes conditional publication if capability and predicates permit.

```
Import and validate -> accept or retain draft -> pin -> evidence checks
   invalid draft: field help                        |
                     calculate -> investigate -> render and hash
                              -> independent exact review
                                   | revise or gaps: new material/review
                                   | approve
                          conditional canonical publication
                                   |
                    current assessment or reasoned refusal
Canonical capability absent: reviewed historical candidate only.
```

A review finding may trigger retrieval or a new input revision. Re-pin and re-review within original time/rounds, or create a linked job. Current status is as-of an authoritative check; source-triggered updates do not mutate old approved artifacts. Gap/no-improvement/threshold outcomes are useful truthful responses, not failures to generate a winner.

The workspace has structured input proposals, baseline/candidate schedules, per-worker and matched-activity results, decision-value/disruption panel, evidence/review drawer and execution/status trace. Separate proposed edits from accepted bundles. Show locked assignments, complete occupancy, services/controls, costs including unknowns, source issue/native resolution, uncertainty scenarios and every blocker. Keep crew totals alongside the distribution. Explain exactly which worker/activity/scenario rejected an option; no waiver control appears.

Use explicit wording: "A numerical projected reduction was verified"; "The declared action floor was not met"; "Engineering candidate only; workplace usefulness unverified"; or "Best verified plan found within the search budget; global optimum unknown." No-improvement wording names the completed searched set. Partial search without a verified gain says that no improvement was verified before cutoff and further improvement remains unknown. A point estimate is not described as robust merely because it exceeds 10%.

Show execution, evidence/check status, agent execution, actual provider/model, explanation source, review, publication and currency separately. Failed Nemotron execution remains failed alongside a successfully reviewed template result. A successful provider call alone does not establish agent investigation or product value. Evidence opens exact revisions. Disconnected views say historical/as-of; refresh before separate action. Programme screens keep damage accounts and action/delivery/outcome state distinct; disabled modules show not_enabled/not_quantified, with no crew percentage relabeled as damage reduction.

17 Modes packaging and learning

17.1 Modes and historical harness

Mode	Required behavior and limits
live	Actual permitted remote evidence/inference operations with explicit provider policy; the canonical surface additionally requires actual configured World Model commands and approved pinned Zetaris queries. Origin and authority surface remain explicit
cached_live	Actual registered views or labeled historical captured rows, with original source ages and access/applicability limits; the canonical surface additionally requires the actual canonical implementation. No claim of fresh observations
offline_sample	Network disabled, no credentials; same v2 planner/checks, deterministic rendering and separate fixture checker; remote Zetaris, Nightshift/NVIDIA and H-MEM explicitly skipped

SAMPLE_MODE enforces offline_sample. No source can be promoted from synthetic to operating by a live query. No offline replay becomes a live environmental assessment. If actual local canonical implementation is absent/unproved, the historical harness uses demo-local immutable files and independent engineering review with current publication disabled. A vendor-named class or local reference protocol is not an installed engine. After canonical proof, a technically current synthetic assessment still remains fictional and ineligible for operational recommendation.

Execution mode and execution_surface are orthogonal. A historical harness may use live Nemotron inference or permitted read-only remote queries on exact demonstration artifacts while canonical writes/current reads remain disabled. Record actual remote operations and unavailable engines independently; a successful remote model call cannot upgrade demo-local references to canonical ones. Only offline_sample forbids the remote operations and forces the template. This permits an honest live model demonstration even while the canonical gate is unproved.

Unseen supported fixtures must be imported/validated and computed from bytes with new actual demo-local or canonical references as appropriate. Never return stored answers or prompt-match expected outputs. Fixtures retain origin, licence, units, native resolution, intervals, hashes and regeneration instructions. Synthetic guidance, attestations, uncertainty and thresholds are fictional. Public fixtures contain no personal or clinical data.

A separate sample run after live failure has its own identities, bundle, trace, deadline and review, linked to that failure; it cannot overwrite the failure or prove a live integration. Queued remote memory capture from offline runs is not marked synchronized or replayed automatically into live memory. A later authorized export/consumer operation has its own scope/key. Optional actual local memory requires the same canonical validity filtering; otherwise skip it.

17.2 Reproducible packaging

Pin implementation/spec commits, package lockfiles, schemas, methods, renderer templates, view definitions and code/checker versions in an engine manifest. Inspect actual starter contents and licences rather than assuming installed capabilities. Build dependencies and actual implementations into the image; no runtime clone, install, model-weight download or manual source preparation. Nightshift remains a separately identified reachable serving dependency.

Startup creates/migrates configured state namespaces and idempotently seeds fictional bundles only where the actual implementation permits. World Model owns canonical state/events; LangGraph owns execution checkpoints; memory and query projections keep their own derived namespaces. Back up and restore with head/event consistency. Migrations cannot rewrite immutable material bytes; new versions retain provenance. If only the historical harness is available, its state is labeled accordingly and cannot masquerade as canonical storage.

Target commands remain specification examples, not executed build evidence:

```
docker build -t haze-harness .
docker run --rm -p 8000:8000 --env-file .env -e STATE_DIR=/data
  -v haze-state:/data haze-harness
docker run --rm --network none -e SAMPLE_MODE=true haze-harness
  node run.mjs --input input_examples/example_v2.json
```

The wrapped command lines must be joined or given the shell's continuation syntax in the implementation README. Commit placeholder-only .env.example, exclude secrets from build/history/assets and provide explicit local principal setup. Document API/CLI, modes, authority, canonical gate, fixtures, schemas/migration, exact checks/results and unresolved prerequisites. Advertise only implemented endpoints. Capture at least comparison, blocked and update/recovery generated runs plus unseen inputs when the application exists; these remain unexecuted requirements here.

18 Evaluation and release acceptance

18.1 Frozen clean and corrupted cases

Before tuning the agent, freeze a versioned held-out subset with input/evidence revisions, as-of cutoff, oracle facts/caveats, expected exclusions/abstention, publication eligibility and permitted wording. Keep separate development cases. Memory contains only prior reviewed material available at test time, with no held-out answers or future evidence. Human usefulness assessment requires an actual qualified participant and comparable information; absent participation is not_run.

Test clean cases and deliberately corrupted artifacts: altered values/units, nonexistent fact IDs, wrong source revision/hash, missing intervals, falsely fine resolution, unsupported indoor assumptions, invalid operating baseline, burden violations, altered scenario definitions, pending/expired reviews, provider provenance mismatch, stale explanation and unauthorized review/publication. Freeze some cases before repair prompts are tuned. Corruptions must be seeded transparently in test environments, not presented as real disagreements.

Measure	Operational definition
Correct explanations	Proportion of cases answering required question dimensions with oracle-supported facts, correct rejection/selection reason and all required caveats
Missed caveats	Missing required caveats divided by triggered caveats, plus case-level count; failures cannot be hidden by terse output
Unsupported claims	Assertions lacking an allowed checked fact/status mapping or exceeding permitted use, divided by material assertions; also report any affected case
False rejections	Clean valid fact selections/plans incorrectly blocked divided by clean valid opportunities; distinguish justified abstention from false block
Fault detection	Corrupted cases correctly detected/abstained, missed faults and false positives by corruption type
Latency and completion	p50/p95 and maximum stage/root latency, sample size, deadline completion, tool/attempt/revision counts and timeouts
Cost	Actual/unknown model, query, compute and memory costs, tokens where supplied, readiness versus admitted versus consumer cost; unknowns are not zero
Product usefulness	Supervisor task accuracy, comprehension, time/effort, missing-fact resolution and decision value under prespecified thresholds; not inferred from a correct model call

Report all denominators and case outcomes. A small four-day engineering suite is exploratory, not statistical proof. Compare planner+deterministic template against planner+Nemotron investigation under identical evidence, actions, constraints and total budgets. Ablate retrieval, critic and memory only when implemented. An agent cannot improve an exact deterministic optimum for identical inputs/objective; its contribution is explanation, evidence handling or reduced user error/effort. Do not retain optional complexity merely because it can be named.

18.2 Required actual application checks

Numerical/operating tests cover independent native-interval arithmetic, full occupancy, fixed identity invariants, breaks/travel/setup, resources, cycles/unknown IDs, zero/negative/nonfinite inputs, unit/time alignment, missing coverage, matched worker/activity violations in each scenario, floor boundaries, movement/count/caps and deterministic tuple ties. Include a case where worker total improves but one matched activity worsens. Check complete-set enumeration/certificates separately from incumbent arithmetic. Test incomplete search both with and without a verified improving witness and target attainable versus unknown. Also complete producer enumeration while leaving an improving proposal unverified; require improvement_not_established and unknown absence/target feasibility, never a false no-improvement conclusion.

Claim/agent tests check each type's verifier, exact locators, mandatory caveats, unsupported science quarantine, grounded questions, actual tool investigation, empty/altered IDs/values, failed evidence tools, valid clean selections and false rejection rates. A schema or hash pass must not bypass a failed check or pending applicability. Post-review alteration of any material value/claim/provider label requires new material/review; presentation-only changes preserve exact content.

Authority tests use the actual canonical implementation: concurrent accept/hold/withdraw versus publish, paused consumers, relevant/unrelated/unseen/ambiguous relevance, routing barriers, lagging dependency projections, expiry, revoked access, self-review denial, conflicting/duplicate keys and authoritative current reads. A local reference pass cannot clear the actual-adapter gate. Any stale/unauthorized publication, silent worker/activity increase or unsupported material claim is release-blocking, not averaged away.

Recovery tests interrupt before commit, after commit before acknowledgement, before checkpoint persistence, during projection/memory work and after lease/deadline expiry. Verify original command reconciliation, no duplicate logical effects, immutable history, durable pending events, retries/dead letters, backup/restore and original budgets. Required provider tests include healthy HTTP but failed generation, token/discovery failure, wrong/unverified route, malformed selection, slow/unknown outcome, cancellation and declared template fallback. Test actual feature provenance separately from prewarm.

Package checks cover clean build/start, authenticated API/CLI, network-disabled no-credential unseen input, actual loaded engine identities/licences and genuine operation receipts. Product checks inspect distribution, activity rejections, source resolution, caveats, unknown costs and current/as-of wording. Field/scientific checks require their own protocols and evidence. All application, provider, usability, concurrency and field tests in this section are requirements until captured; this document revision does not execute them.

18.3 Learning and comparative field evidence

Keep ordinary schedule comparator, software comparator and causal field comparator distinct. Engineering replay checks information handling; shadow planning checks practical use; separately authorized field comparison estimates delivered effects. Neither replay nor shadow work proves avoided actual exposure or harm. A before/after decline alone is not causal attribution.

The experiment lifecycle is hypothesis, prespecified protocol, independent design review, authorized simulation/pilot, implementation record, observations, comparative estimate, independent result review and accepted learning. World Model owns these records; H-MEM receives only eligible reviewed learning. Record primary outcome, population/place/time, resources, measurement/analysis, stopping criteria, approvals, deviations and missingness. Freeze methods or reprocess comparison arms consistently after revision. Report inconclusive/rejected findings as well as favorable ones.

19 Four remaining development days

Use four relative development days; do not infer an event deadline or add elapsed days. The sequence assumes parallel ownership of numerical/contracts, state/integration and agent/UI/evaluation work. If fewer people or access are available, reduce optional work and retain a truthful historical harness. The critical dependencies are contract freeze -> independent calculation/fact checks -> investigated/rendered material -> exact review -> publication only after actual canonical proof.

Day	Dependent work and parallel work	Exit evidence and cut line
1	Numerical lead freezes fixed-work v2 contract, value/protection/scenario policy and fixture; state lead probes actual canonical accept/hold/publish/read/event and Zetaris coverage capability; agent/evaluation lead freezes held-out cases and tool/fact/render contracts. Delivery owner seeks supervisor/source/domain evidence and records the real-world gate in parallel	End-of-day real-use finding or workplace usefulness unverified; written canonical capability decision. If no credible implementation path is proved, freeze publication_disabled historical delivery. Do not spend remaining days building a shadow authority
2	Implement/verify input validator, independent interval/checker, worker/activity scenario screening, threshold/movement ranking and search axes. In parallel wire minimal actual adapters if evidenced; build fact verifiers, controlled renderer and read-only tools against the same immutable artifacts	Reproducible fixture arithmetic, clean/corrupted protections, missing/zero/cutoff outcomes and claim records. If provider access absent, retain explicit template path; live Nemotron execution remains unproved
3	Integrate bounded WorkerShield investigation, provenance, exact material hashing and independent review. State owner runs actual adapter races/recovery if viable; UI/evaluation owner exposes all axes and runs frozen clean/corrupted cases	End-to-end reviewed candidate plus evidence gaps and incomplete incumbent paths. Current publication stays disabled unless actual canonical race/read/recovery gates pass. Cut optional memory, extra roles/providers and cosmetic UI before weakening core checks
4	Repair release blockers; rerun affected tests; inspect unseen-input, network-disabled, retry/recovery, current-read and fallback paths as applicable. Capture genuine provider/engine traces where available; package source, results, limitations and demonstration	Honest achieved-gate inventory and reproducible engineering release. If runtime/reviewer cannot complete safely, ship narrower historical results and failed/not_run evidence. No field-effect or scientific-validation claim

Day-one employer/source/domain work is not a promise that participation or approval will arrive. Later real evidence triggers a new accepted gate/bundle and re-review if time permits; it never retroactively validates the fixture. Current publication must also be cut if a later failure invalidates earlier capability evidence. Source/query availability and science remain independent from canonical proof.

Immediate cut lines exclude regional damage implementation, broad paper/fire monitoring, field-effect evaluation, source-prevention dispatch, general peer protocols, new controls/reassignment/overtime, optional profiles, extra lanes/providers and automatic memory synthesis. Retain the chosen provider architecture and honest status even if access is missing. H-MEM execution is optional for the core slice; any verified event obligation is separately documented rather than implied satisfied.

v0.12 records conflicting event dates and unresolved guide/rules wording. This version replaces its old delivery sequence completely. The user-supplied four-day remaining period is the planning constraint, not verification of

the organizer's exact submission calendar or acceptance. The submission owner must obtain current authoritative rules/form and permitted-prior-work evidence separately.

20 Staged damage assessment and accounting

20.1 Programme model and health boundaries

The enduring programme maintains reviewable haze-damage assessment and helps accountable people investigate and evaluate harm-reduction actions. Before activating regional modeling, adopt fire sources, geography, population, exposure period, consequence horizon and a no-target-fire counterfactual. Require a reviewed transport/source-apportionment method and suitable receptor observations. Distinguish concentration, air-quality index, satellite aerosol, hotspots, burned area and population harm.

Compare concentrations with and without specified fire emissions while holding weather and other emissions comparable. Nonlinear source-removal contributions require a declared allocation rule before aggregation.

```
DeltaC(i,t) = C_actual(i,t) - C_no_target_fire(i,t)
population_exposure = sum_i,t P(i,t) * DeltaC(i,t) * delta_t
```

Time in days gives person * ug/m3 * day. Preserve distributions, background concentrations, coverage and uncertainty. Select supported health endpoints before valuation. Match pollutant, endpoint, age/population, geography, concentration range, averaging period/lag and incidence convention; document period conversions, seasonality and any transfer of all-source PM2.5 evidence to smoke. The alternative statistical forms retained from the source are:

```
Using actual-exposure incidence:
DeltaN = sum_i,a,t P(i,a,t) * r_actual(i,a,t)
         * [1 - RR_a(C_cf(i,t)) / RR_a(C_actual(i,t))]
Using counterfactual-exposure incidence:
DeltaN = sum_i,a,t P(i,a,t) * r_cf(i,a,t)
         * [RR_a(C_actual(i,t)) / RR_a(C_cf(i,t)) - 1]
```

These rate conventions cannot be exchanged. Evaluate nonlinear risks at both total concentrations. They describe expected statistical population outcomes, not identified individual deaths or diagnoses. No function, incidence input or scientific transfer is selected or approved by this revision. Acute and long-term estimates remain separate alternatives where endpoints overlap; a short episode is not sustained lifetime exposure. Recurring exposure requires trajectory, lag and follow-up horizon. Other consequences need supported attribution and incremental quantities, including substitution/catch-up.

The crew planner uses none of these health or monetary conversions. It cannot populate a health-damage account by multiplying a crew proxy reduction by an assumed benefit factor. The separately cited Haze Damage Account was not supplied; no additional method is presumed reviewed.

20.2 Linked accounts and overlap controls

Account	Content and aggregation rule
Physical outcomes	Source-attributable exposure, supported health/time/output and other consequences in their own units
Financial incidence	Expenditure, receipts, income and financing by payer/recipient, including transfers
Selected social welfare	Compatible nonoverlapping resource opportunity costs and nonmarket values within an adopted boundary

Never add these headline totals. Each effect has endpoint ID, account lens, overlap group, selected-total flag and exclusion reason. Preserve unpriced/omitted outcomes. Choose one primary mortality framework; exclude all-cause/cause-specific duplication and alternative estimates of the same endpoint. Use VSL or a compatible life-year/DALY route for the same outcomes, not both. Do not add lifetime earnings to comprehensive mortality-risk welfare values; check treatment/lost-work overlap in illness valuations.

Reconcile wages, revenue and net production loss; assets and capitalized income; ecosystem assets/services; carbon stocks/emissions. Transfers stay in incidence without a second resource loss. Pair incremental protection resources with residual harm under the same behavior, not an incompatible unprotected scenario. Declare unit values/components, currency/price year, valuation date, conversion/transfer basis, horizon and discount convention. Broader direct-fire/ecosystem/climate scope needs its own reconciliation; a policy carbon price is not automatically damage value.

Separate parameter uncertainty, structural alternatives, intervention scenarios and omitted categories. Propagate supported uncertainty with material correlations; analyst sensitivities are not probability intervals. Publish covered subtotals, unmonetized outcomes and missing categories together; missing harm is not zero. Completeness needs explicit coverage review.

Historical source-removal harm is different from achievable intervention benefit. Compare expected damage without and with an actual action, incorporating effectiveness, adoption, delivery, maintenance, persistence, displacement/spillovers and distribution. Complete removal of fires is not one intervention's benefit. Report incremental costs separately. A ledger record binds scope, comparator, endpoint/unit, account/overlap rules, monetary basis where justified, exact models/parameters/evidence, uncertainty, coverage and review. Persistence does not validate the science.

21 Workplace activation prevention and accountable delivery

21.1 Operational planning and later field evaluation

Operational planning needs a named employer/supervisor, credible baseline, genuine timing authority/alternative, worker communication/governance, privacy terms and resources. Appropriate occupational-hygiene review must approve exposure specification, source/worksite transfer, native resolution, activity safeguard interpretation and supported uncertainty. Owner/domain-approved action thresholds and costs must be usable. Actual canonical/independent-review gates must pass. None is asserted obtained here.

A separately authorized field comparison additionally needs suitable personal/activity or microenvironment measurements, placement/calibration/co-location, coverage/missingness/analysis, exact plan authorization, implementation/deviation records and independent delivery verification. Determine sample size, duration and

detectable effects through appropriate design analysis, not the two-worker fixture. Track adherence, service quality, disruption, weather/background pollution, concurrent changes, displaced work and full costs. A regional forecast is not a personal monitor.

Projected proxy, actual delivered timing, measured quantities, comparative exposure effect and health/damage effect are distinct. Actual deviations cannot be called compliant with the modeled safeguard without checking what occurred. No safe-duration, protection-effectiveness or health-benefit inference follows from a planning calculation. Field scaling requires evaluated usefulness, effects, capacity and transferability; four development days do not establish them.

21.2 Prevention pathway and later profiles

Source prevention remains a staged ambition. Select a prospective landscape, plausible controllable mechanism and operator with practical authority before a pilot. Candidate questions may include peatland water-management inspection/repair or fire-free land preparation where supported; these are investigations, not endorsed effective interventions. Assess hydrology, land use/rights, access, livelihoods, maintenance, resources and applicable evidence.

The pathway is mechanism/risk evidence -> partner/authority -> reviewed action/comparison contract -> resources/authorization -> implementation -> independent verification -> comparative fire/emissions/exposure/damage evaluation. Detection alone cannot establish responsibility, preventability or downwind damage. Missing detection is not proof of no fire. Hotspot overlap with peat is not a confirmed peat fire.

PeatFire Sentinel remains a disabled later profile for signal investigation. HazeShield later requires verified stocks, operating facility capacity, transport, needs, fairness/budget/access and delivery authority; a mapped building is not an available cleaner-air facility. HazeInfo and CommunityAlert later require exact reviewed audience/place/time content, recipient/language permissions, preview, durable idempotent delivery/correction receipts and expiry. Changed support invalidates dependent unsent messages. No initial public sending, procurement or dispatch is enabled.

A general peer protocol is deferred. If later enabled, messages carry root/task/correlation IDs, capability versions, objective/schema, exact snapshot, tools/permissions, deadline/reservation and operation key; replies are immutable evidence-linked artifacts. Waiting releases slots; cycles/repeated unresolved requests/disabled targets are rejected. Messages cannot change authority, deadlines or accepted state, and do not establish A2A conformance.

21.3 Action contract and continuous operation

Later action records bind exact proposal, target/scenario/evidence/method revisions, operator/practical rights, resources/full costs/funder/capacity, expected additional benefit/comparator/uncertainty, exact review and authorization conditions/expiry, delivery/deviations, verification and comparative evaluation. Lifecycle is proposed, reviewed, authorized, in_progress, reported_complete, verified_complete, evaluated, with cancelled/failed branches. A separate scoped action_hold blocks continuation while preserving lifecycle stage; only authorized exact-revision clearance removes it. Payment, approval, reported work, verified completion and avoided harm remain different facts.

World Model owns actions/authority/delivery. Runtime enforces conditional effects; LangGraph coordinates bounded jobs. Later execution connectors need their own idempotency, receipts, cancellation/compensation and authority contracts. Current planning publication is informational and supplies no purchase, fund-release, roster-write, dispatch or message permission. Keep immutable links to what was authorized and actually done even after later supersession.

Continuous programme operation means configured recurring checks and meaningful bounded jobs, not an endless model loop. Persist cursors/frontiers, queue/coalescing keys, priorities, leases, retry/dead-letter state and review backlogs. Bound backfill/queue growth and reconcile outages. Monitor source age/coverage, routing/projection lag, canonical conflicts, current-read refusals, unsupported claims, burden violations, incomplete search, review expiry, provider provenance, budgets and cost. Memory lag is visible without undoing valid publication.

Pause/resume does not clear holds or reset deadlines. Manual ticks are not described as autonomous monitoring until scheduling/catch-up is executed. Broad monitoring and regional damage models remain disabled until their own access, review, workload and evaluation gates pass. Programme target success requires verified delivery, credible comparative outcomes, reach/adoption/persistence and an adopted uncertainty/attribution criterion; hypothetical reach-times-effect arithmetic is not a Haze forecast.

22 Unresolved prerequisites and owners

Accountable role	Evidence still required	Boundary while pending
Supervisor and delivery owner	Reviewed ordinary schedule, real permissible alternative, timing rights/services/constraints and day-one gate	Engineering demonstration; workplace usefulness unverified
Evidence steward and occupational hygienist	Decision-time numerical access/licence, quality, location/native-resolution support, alignment, uncertainty and intended-use review	No supported operational comparison/recommendation
Policy owner and domain reviewer	Justified meaningful benefit, movement/count/individual caps, costs, uncertainty and fixed-work protections	Numerical gain does not imply action-worthy or operationally eligible
Canonical technical owner	Actual pinned backend, serialized accept/hold/publish/read, durable events, dependency attestations, races/recovery/restore	Current publication/read disabled; historical reviewed candidates only
Query/integration owner	Approved Zetaris views, exact rows/revisions, independent critic queries and verifiable dependency coverage	Affected publication/read blocked; global watermark insufficient
Workflow/review owner	Executable closed schemas, separate checker, fact verifiers/renderer, independent privileges and exact review capacity	Specified behavior only; no self-approval or test-pass inference
Runtime/provider owner	Reachable installed adapter, credentials separation, discovery/generation and concrete per-feature Nemotron provenance	Required live admission blocked or declared visible template degradation
Learning owner	Actual H-MEM capture/recall/audit/supersession and canonical stale filtering	Skip/fail/lag visibly; no fabricated memory contribution
Evaluation/release owner	Frozen cases, captured app checks, latency/cost/user results, reproducible packaging and genuine traces	Contribution and app acceptance unexecuted until evidence exists
Submission owner	Current official rules/form, eligibility/prior work, Power/Build/award wording and exact calendar	No organizer acceptance or verified submission-deadline claim
Employer/field evaluation lead	Privacy/retention, participation/communication, authorized protocol, suitable measurements, verified delivery/comparison	Projection only; no measured effect or health claim
Programme owner/domain panel	Adopted damage metric and scope; review of attribution, health, accounting, valuation, coverage and causality	Regional damage and prevention effects unquantified/unvalidated

The worker and matched-activity no-increase rules are adopted design constraints, not optional unresolved run-time discretion. Relaxation requires a separate policy/contract and domain/worker-governance review outside

this build. No architecture revision resolves unproved backend, access, science or organizer requirements by declaring them done.

23 Fictional engineering example

23.1 Inputs and independent arithmetic

All workers, locations, services, rights, concentrations, uncertainty scenarios and thresholds here are fictional. The example tests mechanics, not Malaysian observations, employer practice, forecast accuracy, scientific approval, dose, safety or field benefit. The window is 08:00-12:00 on 25 October 2026 in Asia/Kuala_Lumpur. Two workers share the same three canonical activities: inspection for one hour, maintenance for one hour after inspection, and administration for two hours. Locations are fixed to yard for inspection/maintenance and office for administration. Fictional constraints attest no travel/setup requirement for this example only. Both workers have complete occupancy.

Native hourly interval	Yard PM2.5 ug/m3	Office PM2.5 ug/m3
08:00-09:00	100	20
09:00-10:00	80	20
10:00-11:00	50	20
11:00-12:00	30	20

The office series is independently invented, not a filtration ratio. A 15-minute planning grid uses piecewise-constant hourly information. A second fictional sensitivity scenario adds 20 ug/m3 to every yard and office hour. It is an engineering perturbation, not a supported forecast-error distribution or operational uncertainty treatment.

Declare a fictional action floor of at least 10% crew reduction in both scenarios, total movement cap 360 activity-minutes, changed-activity cap 3 and each activity movement cap 120 minutes. These are fictional engineering choices, not owner/domain-approved operational limits. Shared activities are counted once for movement.

Plan and starts	Per worker nominal proxy	Crew nominal proxy	Total movement M and count N
Baseline I08 M09 A10	220	440	0 minutes and 0
Less disruption I08 A09 M11	170	340	180 minutes and 2
Lower proxy A08 I10 M11	120	240	360 minutes and 3

I, M and A mean inspection, maintenance and administration. For each worker the baseline matched-activity integrals are I=100, M=80, A=40. The less-disruption plan gives I=100, M=30, A=40; the lower-proxy plan gives I=50, M=30, A=40. All nominal matched activities and worker totals do not increase. Baseline is 100+80+40=220 per worker; less disruption is 100+30+40=170; lower proxy is 50+30+40=120. Crew nominal reductions are 100/440=22.73% and 200/440=45.45%.

In the +20 scenario each worker's proxy increases by 80 from four hours, so baseline=300, less disruption=250 and lower proxy=200; crew values are 600, 500 and 400. Scenario reductions are 16.67% and 33.33%. Matched-activity values are baseline 120/100/80, less disruption 120/50/80 and lower proxy 70/50/80. Both candidates pass fictional benefit, movement and protection filters. The v0.13 policy selects the 340 nominal crew plan because 180 minutes is less movement than 360. It does not select the 240 plan merely because its proxy is lowest.

Movement arithmetic uses baseline starts I08/M09/A10. The selected candidate moves M by 120 and A by 60 minutes, totaling 180 with two changed activities; the lower-proxy option moves all three by 120, totaling 360. No per-worker double counting occurs. Per-worker units remain hours * ug/m3 and crew units person-hours * ug/m3.

23.2 Claims and search variants

Checked calculated_difference facts can render the nominal/scenario values; decision_policy_finding explains selection; source_descriptor states fictional hourly information; uncertainty_finding states only the fictional scenario outcome; pending_review states operational applicability unresolved. WorkerShield must inspect the actual comparison and a relevant policy/source/sensitivity dimension. A copied example paragraph is not successful agent execution. Exact provider/tool provenance and the final rendered facts precede review.

If search stops after finding and independently checking the 340 candidate, retain its verified gain and fictional threshold witness with search_status=incomplete and optimality_status=unknown. It may be the selected engineering candidate with the budget-limited wording if best among independently verified plans. target_feasibility=attainable for that fictional target; recommendation_eligible=false because inputs are synthetic and operational/canonical gates are unresolved. Do not assert global preference over undiscovered plans.

If search stops without a verified improving candidate, retain baseline and say no improvement verified within budget; target_feasibility=unknown. If yard/office trajectories are flat at each fixed location, fixed durations cannot create a gain by reordering labels. Missing baseline hourly coverage gives no quantified comparison. A modified fixture with worker total falling from 100 to 95 but matched activities changing from 80/20 to 70/25 is rejected: the second activity worsens by 5 even though total improves. These variants test distinct semantics.

The original reference package and its reported 1,521 combinations, three feasible plans and 21 prior checks were not supplied or rerun here. This revision independently recomputes the displayed example arithmetic and policy comparisons in a document-check script; it does not execute the old enumerator or Haze application. The displayed three schedules are illustrative, not a newly proved exhaustive feasible set. A global optimality assertion requires actual search/certificate evidence.

24 Source register and reference limits

24.1 Supplied source and inherited provenance

[I6] Haze_Agent_Architecture_v0.12.pdf, 56 pages, 23 October 2026, is the sole supplied architecture source for this revision. Its complete text, tables, diagrams, references and appendices were read, with all pages visually inspected. SHA-256: c7ee9bacc3075155fb6b82e175fd9159737df20be4fb2a55aa2be3868c8184b1. The original download remains unchanged.

Retain the following identifiers as inherited provenance from I6, not newly supplied or inspected companion files:

Identifier	Recorded source and limitation
I1	Haze_Agent_Architecture_v0.10.pdf, 28 pages; earlier programme mission/accounts/expansion source reported in v0.11/v0.12
I2	Haze_Five_Changes_Implementation_Blueprint.pdf, 8 pages; earlier proposal reported by the source, not a separate authority here
I3	Haze_Five_Changes_Reference_Package.zip; reported synthetic enumerator/SQLite illustration and prior 21 checks, not newly executed
I4	Haze_Agent_Architecture_v0.11.pdf, 35 pages; authoritative baseline for v0.12, not supplied to v0.13
I5	Open Agent Hackathon 2026 Participant Setup Guide.pdf, 30 pages, recorded update 19 October 2026; integration wording reported by v0.12, not current rules or installed capability evidence

I6 records I4 SHA-256 as 1c9144f575433c1c0e9f2e9ce89561a6e724db0cfbfdd88f93c003739d410be8 and I5 as 099b2eb6e70cd1dca5363d1ec60fca1918ce949edd44377464ed879c26afec61. Those are inherited identity claims, not hashes recomputed from unavailable files. The separately named Haze Damage Account and 118-page Technical Execution Requirements compilation were not supplied; their contents are not assumed reviewed.

Domain leads retained for later retrieval include METMalaysia numerical products, NASA FIRMS notes, applicable DOSH/Ministry of Health haze guidance, WHO AirQ+, indoor exposure evidence, emissions/transport products and original peat-restoration/political-economy studies. They are leads, not accepted parameters. Before activation obtain exact official revision/URL/DOI, licence/access, raw hash, coverage/resolution, method/quality evidence and qualified permitted-use review. No medical, legal, epidemiological, physiological or monetary parameter is newly adopted here.

24.2 Retained technical references

These links and their limited descriptions are retained from I6's account of earlier checks. They are not newly verified external requirements in this document. Installed code versions, actual concurrency and source applicability still require direct evidence. References support mechanisms to investigate, not a claim that Haze passed them.

Ref	Reference and bounded relevance
T1	https://json-schema.org/understanding-json-schema/reference/object - object/required/closed structure; not cross-record truth or feasibility
T2	https://docs.langchain.com/oss/javascript/langgraph/persistence - execution persistence concepts; not cross-engine canonical transactions
T3	https://docs.langchain.com/oss/javascript/langgraph/fault-tolerance - configurable retry/timeout mechanisms; pin actual version and shared runtime limits
T4	https://github.com/Meterless/World-Model - source-recorded implementation specification; no deployed conditional command capability inferred
T5	https://github.com/Meterless/H-MEM - source-recorded specification/provenance concepts; actual runtime operations unproved
T6	https://academy.genai.works/courses/open-agent-hackathon-2026/details - source-recorded organizer course page; not complete current rules/form
T7	https://github.com/ryprojects1/OPENAGENTHACKATHON2026 - source-recorded rules mirror with unresolved authority/date discrepancy; not adopted event contract
T8	https://www.sqlite.org/lang_transaction.html - one-database transaction mechanisms; no automatic transfer to World Model, views, memory or checkpoints

Before implementation claims, pin actual packages/backend/spec revisions and inspect the relevant APIs/licences. Before organizer claims, retrieve authoritative current rules/form. Reading an instruction in a source never authorizes external configuration, messages, deployment or field work. This revision performs document work and explicitly scoped arithmetic/layout checks only.

Appendix A Material changes and preserved guarantees

v0.12 area	Integrated v0.13 change and rationale
Mission/activation and former delivery calendar	Section 1 makes day-one real decision usefulness testable, with explicit absent-flexibility/information/value outcomes; section 19 replaces the old sequence with four remaining days and cut lines
Metrics/burden/uncertainty	Sections 4-6 retain worker totals and add matched worker/activity checks in every required scenario, with limited concentration-time interpretation and occupational-hygiene gate
Ranking and target	Sections 6-7 apply justified benefit/disruption policy before ranking; witness target attainment is separate from search-wide proof and operational eligibility
Planner/outcome/UI/API	Sections 7 and 15-16 remove search-complete-only improvement semantics; an independently checked incumbent survives cutoff with explicit optimum limits
Explanation/agent	Sections 8-9 replace broad semantic faithfulness with typed facts, per-type checks and controlled rendering; WorkerShield performs bounded record investigation
Contracts	Section 15 versions changed planning/run/result/agent meanings to v2 and requires explicit migration; inference receipt binding stays v1
Review/canonical/recovery	Sections 10-13 retain exact material-before-review, acyclic provenance binding, epochs, holds/routing, authoritative current reads, permissions, idempotency, fencing and recovery; no unproved engine capability is assumed
Provider/modes/evaluation	Sections 14 and 17-18 preserve Nightshift/Nemotron and honest template fallback, add actual investigation and frozen clean/corrupted evaluation; execution/usefulness/acceptance remain distinct
Programme and field scope	Sections 20-22 retain attribution/accounts/overlap, prevention ambition, action/delivery and causal evaluation as staged capabilities; no immediate regional or field-effect build
Example and source provenance	Sections 23-24 recalculate fictional decision-value selection, preserve source limitations and identify the only supplied PDF without claiming companion inspections/tests

Appendix B Second complete scenario review

This is a second documentary trace after the integrated revision, not captured application execution. Each path follows inputs, authority, calculations, agent, evidence checks, exact review, publication and visible output. Actual application tests remain section 18 requirements. References below are section references, not assertions that unavailable capabilities were executed.

B1 Useful supported improvement

Inputs are a supervisor-reviewed ordinary schedule and permissible equivalent-work alternative, complete v2 operating facts and predecision suitable evidence (1, 3-4). Authorized stewards accept the current bundle/policy; source documents grant no authority (2, 10-11). Planner and separate checker reproduce full occupancy, integrals, worker/activity scenario nonincrease, benefit floor and disruption selection (5-7). WorkerShield actually inspects comparison plus relevant evidence/rejection/sensitivity, selects passed facts and grounded dependencies; renderer adds mandatory caveats (8-9). Identity/alignment/numbers and qualified intended-use review remain distinct checks (4). Exact material and provenance precede independent approve (10-11). Only a proved canonical gate can publish current; recommendation_eligible additionally requires all operational gates, not complete search (11, 15). UI shows verified projected benefit, movement, distribution, source resolution, uncertainty and optimum scope; no roster action or health guarantee (16, 21).

B2 Missing or unsuitable evidence

An accepted operating bundle can explicitly record environmental gaps; missing operating facts instead follow B3 (3-4). Pinned checks find exact baseline/location/scenario gaps, issue-time failures or missing applicability for intended use. No absent value becomes zero and no agent fills it (4-5, 8). Essential baseline/use insufficiency prevents quantified comparison; candidate-only gaps exclude those alternatives and restrict scope (7). WorkerShield investigates coverage/review status, renders passed missing_coverage/pending_review facts and asks grounded next-information questions (8-9). A separately complete gap-only candidate receives exact independent review; insufficient numerical material cannot publish (11). If canonical predicates pass, a current gap assessment may describe the exact hold with no selected plan/quantified benefit and eligibility=false; otherwise historical only (11, 15). UI names what is missing/unsuitable and workplace usefulness unverified. Later valid evidence requires new acceptance/epoch, calculation and review (12-13).

B3 Invalid operating inputs

Unknown IDs, cycles, infeasible ordinary baseline, missing occupancy/services, unsupported applicable constraints or unattested fields are caught during controlled import or admission revalidation (3, 15). Only authorized acceptance can create a head; a model question cannot accept a correction (2, 9). No valid calculation, fabricated baseline or numerical recommendation is produced. Deterministic help returns field/check-specific questions; the accepted-bundle agent interface is not invoked with invented references (9). Structural and domain errors have 400/422 as appropriate; present but unattested drafts remain incomplete (15). There is no exact decision approval or current publication for the rejected draft. UI shows validation/help and no selected plan. Once corrected/attested, a new valid bundle and separately admitted run use their own snapshot/budget (10, 13).

B4 No meaningful improvement

Valid operating evidence and full required scenario coverage permit comparison, but a fully checked declared set has no positive qualifying gain, for example flat fixed-location concentration histories (3-7). Authorized policy remains unchanged; no threshold is relaxed to force a winner (6). Checker verifies baseline and search coverage/certificate separately. WorkerShield investigates limiting constraints/evidence and renders the precise no-improvement or no-eligible-plan scope, including all exclusions (8-9). Independent review checks absence wording and search scope, not just a zero percentage (11). A current nonrecommendation assessment can publish only through the canonical gate; baseline retained, selected plan null and eligibility=false (15). UI states no demonstrated meaningful improvement in the named set; uncovered options or incomplete search forbid a global negative finding (7, 16). Day-one gate may record no_meaningful_improvement only with that adequate basis (1).

B5 Improvement below the action threshold

Inputs/policy are valid and predeclared; a checked candidate has positive nominal/scenario gain and passing protections, uncertainty and disruption/cost limits, but fails the benefit floor (3-7). It remains a numerical improvement, with action_threshold_status=below. WorkerShield inspects comparison and policy findings; controlled claims state the exact gain and unmet floor, without calling numerical tolerance meaningful benefit (8-9). Evidence/use checks and exact review cover those statements (4, 11). selected_plan=null and comparison_plan may show that candidate; current assessment publication, if available, is informational and recommendation_eligible=false (15). UI retains baseline, displays the below-floor gain and unknown costs, and makes no instruction to move work (16). Target test is evaluated separately under its own rule (7).

B6 Worker or matched-activity burden violation

Baseline/candidate preserve activity IDs, assignments/durations/locations/controls and full occupancy; every required scenario is paired (3-5). Checker identifies a worker-total increase or a matched activity increase even where crew/worker totals fall. The immutable policy has no supervisor waiver (5-6). Violation facts name worker/activity/scenario and interval arithmetic. WorkerShield inspects rejection and can explain it, but cannot select the excluded plan or invent physiology (8-9). Review independently reproduces both protections and exact claims (11). Another qualifying checked plan may be selected; otherwise baseline is retained and all rejection reasons shown. A complete-set conflict summary requires checked exhaustive scope; partial search describes only inspected rejections (7). The violating plan is never operationally eligible or selected, although a reviewed rejected-option diagnostic may appear in an assessment (15-16).

B7 Incomplete search with verified improving incumbent

Inputs/evidence are valid. Search reaches its 30-second reservation/cutoff after persisting a qualifying incumbent; independent checking verifies its full constraints, scenario protection, benefit and movement (3-7, 13). Search completion stays incomplete and global optimality unknown. It remains a verified positive gain and, if best among independently verified plans, selected_plan may be nonnull. WorkerShield investigates the incumbent, evidence and cutoff scope; renderer states "Best verified plan found within the search budget; global optimum unknown" and the checked-plan count (8-9). Review binds that exact qualified wording and all checks, without demanding an exhaustive search to acknowledge gain (11). A checked target witness may make target_feasibility=attainable; it does not prove optimum (7). Operational recommendation is possible only if every other real-use, policy, uncertainty, review and canonical gate passes; synthetic/unproved-canonical cases remain ineligible (11, 15). UI preserves gain and cutoff together (16).

B8 Incomplete search without verified improvement

Valid baseline survives, but no improving candidate has passed independent verification before cutoff; producer-only gains are not promoted (3-7). Search_status=incomplete, selected_plan=null and target_feasibility=unknown absent a separate checked witness/bound. The result says search_ended_without_verified_improvement, not no improvement exists or target unattainable. WorkerShield inspects available comparison/search/gap facts and renders the uncertainty, with no invented rejection of unseen plans (8-9). Review checks the limited claim and any baseline/comparison arithmetic (11). A current reviewed nonrecommendation assessment may publish through a proved gate; operational eligibility is false (15). UI retains baseline, identifies cutoff and unresolved search space, and may suggest a new separately admitted search without extending the original deadline (13, 16).

The corresponding verification-cutoff variant has producer search_status=complete but search_verification_status=partial and an unchecked improving proposal. It yields improvement_not_established, not an absence claim. Search completion, checked improvement and checked absence remain separate; exact review cannot infer missing checks from the producer's completion flag (7, 18).

B9 Unsupported claims or altered values

The source bundle/check records remain pinned. The model returns a new value, nonexistent fact, omitted mandatory caveat, unsupported science or a false rejection premise (8-9). Selection/renderer checks fail before material creation; a schema pass or cited source does not authorize the assertion. WorkerShield cannot mutate canonical facts. A bounded charged repair or declared checked template may replace the failed draft before hashing, with agent/inference failure visible as applicable (9, 13-14). If alteration occurs after exact review, the material hash differs and publication is refused; new material and independent approval are required (10-11). No post-review free-form assertion is displayed. If no valid reviewed material fits, return an unpublished diagnostic/failure. UI shows truthful failure/fallback rather than the corrupted values (15-16).

B10 Relevant evidence changes after review

An authorized acceptance/hold/withdrawal recognizes exact/subscription relevance and serializes with publication on affected heads, even if consumers are paused (10-12). It advances the epoch or fence and records a durable event. The previously calculated/approved artifact remains immutable historical material. Its expected-epoch/current predicates fail; publication cannot race past the change and read_current cannot serve it as current from a lagging view (11-12). WorkerShield's old facts/rendering are invalid for current use and cannot silently attach new citations. Re-pin, recalculate/check, reinvestigate/render and re-review within remaining time/two shared rounds, or start a linked job (9, 13). UI shows needs_review/withdrawn/stale reason until valid replacement. Unrelated revisions instead preserve this epoch when relevance is proved; ambiguity holds rather than assumes irrelevance (12).

B11 Provider failure and declared fallback

Explicit provider/profile/configuration and independent caller auth precede admission (14-15). Failed required readiness blocks runtime_required with 503 and admitted=false; template-permitted admission is visibly degraded. After admission, wrong/unverified route, timeout or malformed output fails the Nemotron/agent assertion even if arithmetic is valid. Shared attempts/time include tools, backoff and repairs (9, 13-14). The declared deterministic path resolves checked records and mandatory caveats, materializes source=deterministic_template and failure provenance before exact review. No reviewer/check failure is waived. Canonical/operational eligibility is evaluated normally and independently of provider success (11, 15). UI can show completed reviewed numerical response with failed inference/agent evidence, unknown cost where needed and no successful award/integration claim. A new offline run cannot erase the failed live run (17).

B12 Canonical capability unavailable

Technical discovery or actual failure tests cannot establish the required installed conditional command/read/event/coverage boundary (11). No public specification, checkpoint store or reference mutex fills that authority gap. Inputs/calculations/facts may be exercised in a clearly labeled historical harness with demo-local refs and separate engineering review, or the actual implementation's permitted immutable historical path (11, 17). WorkerShield can investigate those exact historical artifacts and explanation provenance remains honest (9, 14). Review scope explicitly says engineering/historical; it cannot approve current capability into existence. Current publication and current-recommendation reads remain disabled in every mode, selected engineering candidates are not operational recommendations, and recommendation_eligible=false (15). UI says publication_disabled and workplace usefulness unverified where real evidence is absent. No shadow canonical store is introduced; releasing a reduced demonstration records the unmet gate (19, 22).

B13 Interruption retry and recovery

Interrupt before/after commit, before acknowledgement/checkpoint, during query/memory consumers or after lease expiry (13, 18). Canonical authority determines whether the keyed effect committed. Recovery first queries the same key/fingerprint; it returns the prior committed outcome or safely retries only while original permissions/lease/deadline permit. Conflicting key reuse rejects. Calculation/fact/render/candidate hashes and exact review remain bound; checkpoints cannot declare a stale result current (10-13). Resume retains original time, attempts, reservations and shared rounds; expired work becomes timed_out and continuation is a newly admitted linked job. Already committed results can be read after expiry, but no new late effect occurs. Durable events replay at least once with idempotent logical consumers, visible lag/dead letters and reconciliation; memory failure does not undo publication. UI retrieves authoritative currency and historical run outcomes, with no duplicate logical action or automatic external dispatch (15-17, 21).

Appendix C Reconciled invariants and review limits

The complete second pass traced sections 1-24 and Appendices A-B against these invariants: one fixed-work contract; complete occupancy; one policy before ranking; worker and matched-activity protections across required scenarios; decision-time evidence; typed checked material assertions; actual bounded agent investigation; explanation completion before exact review; acyclic material/provenance hashes; one canonical authority; synchronous current fences; independent publication/currency/operational eligibility; separate improvement/search/optimality; original budgets through retries; and truthful provider/mode/recovery fallback.

The revision removes the old search-complete-only improvement definition, automatic proxy-first selection, passive stored-explanation agent contract, generic semantic-approval path, implicit v1 compatibility and old delivery sequence. Those obsolete behaviors are not normative anywhere in this version. Tables, example, API fields, target semantics, scenario traces and required tests use the revised meanings. Cross-references identify sections rather than inherited page numbers; the generated contents identifies current PDF pages.

Review findings are documentary: they establish the stated behavior across branches and record prerequisites, not implementation, execution or scientific truth. The displayed fictional arithmetic, budget totals, source preservation hash, generated text/navigation and final rendered pages are checked during document preparation. The old reference suite, actual Haze app, live engines/provider, concurrency/recovery, usability/cost benchmark, employer shadow trial and field/scientific validations remain unexecuted or unavailable for this revision. Their precise evidence owners and disabled boundaries are in section 22 and the companion report.